

**The grammar of gender:
Insights from Bantu asymmetries of AGR with conjoined subjects**
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1. Introduction.

In this paper I argue that asymmetries in agreement with conjoined nominals provide a number of important insights into the nature and workings of grammatical gender and gender agreement. The empirical base is mainly data from the Bantu language Xhosa, with some additions from related Shona. I follow Carstens (1991) among others in analyzing odd/even pairs of noun classes as singular/plural of 5+ genders, as will be detailed in §2; Xhosa has 5 of these and Shona has 8+. The large inventories of genders in Bantu provide information beyond what is available in 2-3 gender languages.

My focus is a set of systematic contrasts found in both Xhosa and Shona involving conjoined singulars and conjoined plurals.

In some genders, conjoined singulars pair with gender-matching plural agreement as shown in the Xhosa (1); the subject marker in (1)a,b is the same as accompanies a plural in (1)c. Three genders exhibit this pattern: those corresponding to noun classes 1/2, 7/8, and 9/10.

- (1) a. Um-mi nom -ongameli **ba**-ya-ncokola. [1+1=2]
1-citizen and.1-president 2SM-DISJ-chat
'The citizen and the president are chatting.'
- b. U-loliwe kunye no-matshini **ba**-ya-hamba. [1+1=2]
1a-train and and.1a -machine 2SM-DISJ-move
'The train and the machine are moving.'
- c. Aba-mi/oo-matshini **b**-a-wa.
2-citizens/2a-machines 2SM-ASP-fall
'The citizens/machines fell.'

With conjoined singulars of other classes including class 5 below, gender-matching plural agreement is not possible. Instead, there are two agreement options in such circumstances:

class 2 for human-denoting conjuncts like (2)a, and class 8 for others as in (2)b. (2)c shows that the class 6 *a-* subject marker which is ruled out in (2)a,b is required with plurals of this gender.

- (2) a. I-gqwetha ne-sela √**ba**/X**a**-phumile. [5+5=2; #6]
 5-lawyer and.5-thief 2SM/ 6SM-left.
 'The lawyer and the thief left.'
- b. I-qanda ne-cepe X**a-**/√**z-a**-wa. [5+5=8; #6]
 5-egg and.5-spoon 6SM/8SM-ASP-fall
 'The egg and the spoon fell.'
- c. Ama-sela X**ba**/√**a**-phum-ile.
 6-thieves 2SM/ 6SM-leave-PST.DISJ
 'The thieves left.'

Xhosa class 3 nouns also exhibit this pattern and in Shona, several classes of singulars do so.

These asymmetries are not Bantu-particular – in Bosnian, Croatian and Serbian (BCS) and Slovenian, feminine and masculine pattern like Xhosa class 1, while neuter gender behaves similarly to Xhosa classes 3 and 5 ((3) from Murphy & Puskar 2018; (4) from Despic 2017).

- (3) [_{&P} Zavesa i biljka] su ukrašavale prozor.
 curtain.F.SG and plant.F.SG are decorate.PRT.F.PL window
 'A curtain and a plant decorate the window.'
- (4) Naše selo i celo jedno brdo su izgorel-i/*-a u požaru.
 our village.NEUT and whole one hill.NEUT AUX.3.PL burned-M.PL/ *NEUT.PL in fire
 'Our village and one whole hill were burned in the fire.'

To account for this pattern Despić (2016), Nevins (2018), Tsimpli & Hulk (2013), Adamson & Anagnostopoulou (2024) propose that neuter is the absence of gender or negative values [-fem, -masc]. Approaches like this do not extend well to Xhosa where there are two such genders and as we will see in §7, Shona has six; they outnumber the genders with the regular plural pattern. It's also claimed to be relevant that neuter contains mostly mass nouns that are non-count (Arsenijevic 2017, 2021) and pattern as defective in certain ways, but classes 3/4 and 5/6 do not have comparable properties.

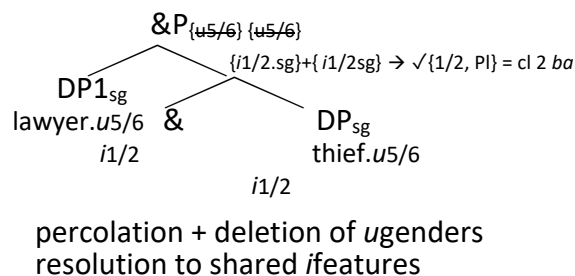
The second contrast is a different kind of asymmetry exhibited by agreement with conjoined gender-mismatched plurals. First conjunct agreement is preferred with DP1 of plural classes 2, 8, or 10 as shown in (5)a; but the preference shifts to last conjunct agreement if DP1 is of classes 4 or 6 as in (5)b,c; these are plurals to the problematical singular classes 3 and 5.

- (5) a. lin-tombi **naba-ntwana** zi/ba-ya-cul-a. [10+2: FCA>LCA 5:3]
 10-girls and.2-children 10_{SM}/2_{SM}-DISJ-sing-FV
 'The young ladies and the children are singing.'
- b. Imi-gewu na-**ba-ntwana** ba/i/zi-ya-cul-a. [LCA>FCA>[-human] default 5:2:1]
 4-criminals and.2-children 2_{SM}/4_{SM}/8_{SM}-DISJ-sing-FV
 'The criminals and the children are singing.'
- c. Ama-polisa na-**ba-ntwana** ba/i-ya-cul-a. [LCA >FCA 6:2]
 6-polisa and.2-children 2_{SM} /6_{SM}-DISJ-sing-FV
 'The policemen and the children are singing.'

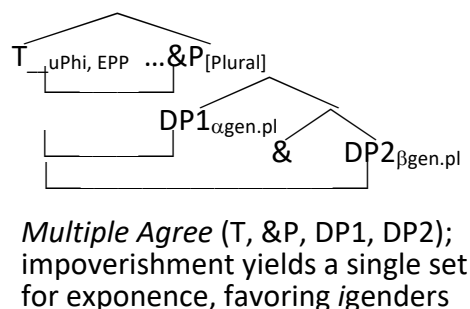
The proposals I will advocate are the following. First, genders like neuter and Xhosa 3/4 and 5/6 have no *i*(nterpretable) content that is legible in narrow syntax; I characterize them as *u*(nterpretable) genders which always stack above *icores* of 1/2, 7/8, or 9/10. Second, for this reason gender resolution on &P deletes them, yielding “default” agreement with conjoined singulars reflecting their *icores* since only genders with *i*content are factored into resolution, adapting a proposal in Adamson & Anagnostopoulou (2024) to predict the patterns of agreement with Xhosa conjoined singulars as sketched in (6)a. Third, genders which have both arbitrary and non-arbitrary members are reasonably taken to have *i*- and *u*flavors (Kramer 2015), but the flavors are not distinguished in the grammar of agreement with conjuncts as (1)a,b show; this is true of Bantu classes 1/2, 7/8 and 9/10 and BCS feminine and masculine. I propose that such flavors are indistinguishable until LF licensing conditions ensure appropriate [gender+root] pairings (adapting Kramer 2015). Fourth, the properties of agreement with conjoined plurals differ from those of conjoined singulars because only singular number conjuncts trigger gender resolution, contra Corbett & Mtenje’s (1987) claim that gender mismatch also does so. But the

facts of conjoined plurals are completely at odds with it; for plurals, a Multiple Agree analysis reaching into &P to copy features of both conjuncts succeeds where the resolution approach fails. These proposals yield the contrasting derivations in (6)a,b).

(6) a. Agreement with conjoined singulars



b. Agreement with conjoined plurals



Fifth, Shona facts suggest that regardless of the number of genders in a given language only 2-3 of them have legible *i*flavors; of Shona's 8+ genders only 2 pattern as having *i*flavors on the diagnostic of gender-matching agreement with conjoined gender-matched singulars.

Last but not least is my claim that the success or failure of agreement with conjoined singulars diagnoses gender interpretability, making possible this set of insights.

This paper has nine sections. §2 presents the basics of Xhosa noun classes and agreement. §3 details the patterns of asymmetry in agreement with conjoined singulars. §4 sketches the ingredients of the analysis, and §5 applies them to derive the patterns. §6 describes and analyzes the asymmetries of conjoined plurals. §7 explores a hypothetical role for non-canonical semantic content in illegibility of genders, applying to Xhosa classes 3/4 and 5/6 some diagnostics from Arsenijevic (2017)'s exploration of BCS neuter. Based on negative results in Xhosa, the hypothesis fails. §7 also shows that in Shona, only two of eight singular/plural paired classes exhibit regular plural agreement with conjoined singulars and concludes that agreement failure is the norm rather than the exception. Among the outliers are diminutives, for which the

facts motivate an account differing from approaches advocated in Kramer (2015), Fuchs & Van der Wal (2022). §8 concludes.

2. Xhosa basics

2.1 Noun class overview

Xhosa has five regular singular/plural pairings exemplified in (7). These pairings are generally identified as noun classes 1-10 following a numbering system that dates to Meinhof (1906) and work of Malcolm Guthrie (1948, 1967, 1971)^{1,2} (for clarity I color-code class prefixes and the agreement that distinguishes classes in the data and/or in glosses).

- (7) a. **um**-ntu/**aba**-ntu b. **um**-pu/**imi**-pu c. **ili**-so/**ame**-hlo d. **isi**-tya/**izi**-tya e. **in**-ja/**izin**-ja
1-/2-person 3-/4-gun 5-/6-eye 7-/8-dish 9-/10-dog
'person/s' 'gun/s' 'eye/s' 'dish/es' 'dog/s'

I ignore three additional Xhosa classes: a class 11 that pluralizes in class 10 (the same as singulars of class 9), and two classes, 14 and 15, which lack plurals; these include mainly abstract nouns, infinitives, gerunds, and deverbal nouns.

A number of researchers have argued that grammatical genders underlie each Bantu singular/plural noun class pairing (see Carstens 1991, 2008, Corbett 1991, Corbett & Mtenje 1987, Fuchs & van der Wal 2022, Watkins 1937). Example (8) shows the system of five genders A-E proposed in Carstens (1991). Carstens proposed that a noun enters the syntax endowed with a gender feature and raises to adjoin to the head of a Num(ber)P projection; updating this to reflect the *n* approach to gender yields (9). In Carstens's system, noun class prefixes are

¹ Examples are from my fieldwork unless otherwise noted. Empirical generalizations regarding conjoined singulars are drawn from Taraldsen et al (2018) who cite 21 Xhosa speakers, and from follow-up field research of my own with 8 speakers, all quite consistent in judgments. Xhosa and Shona examples of conjoined plurals in §6 are from my fieldwork and provide breakdown of responses since there is variability.

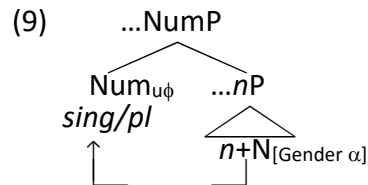
² In glosses, numerals indicate noun class (number+gender); SM = subject marker; OM = object marker; ASP = aspect; DISJ = disjoint, a VP-final verb form; LOC = locative. PST = past tense; PRT = participle; F(em) = feminine; M(asc) = masculine; N(eut) = neuter; PL = plural; POSS= possessor. [X] denotes semantic feature X.

gender-particular spellings out of number features as shown in (10).

(8) Bantu Genders (Carstens 1991)

Gender A: stems of Classes 1/2
 Gender B: stems of Classes 3/4
 Gender C: stems of Classes 5/6

Gender D: stems of Classes 7/8
 Gender E: stems of Classes 9/10
 Etc.



(10) Nouns class prefixes are gender-specific number morphology

[Singular] ↔	um-	/__N _[Gender A]	[Plural] ↔	aba-	/__N _[Gender A]
[Singular] ↔	isi-	/__N _[Gender D]	[Plural] ↔	izi-	/__N _[Gender D] etc.

I adopt Carstens's analysis of noun classes as singulars and plurals of genders A-E, but for convenience I utilize the standard numeric classes in glosses and much of the discussion.

There is abundant agreement in noun class on Xhosa nominal modifiers and predicates.

Table 1 illustrates the subject and object markers and adjectival concord for each class. I include subclasses 1a/2a as I will refer to them from time to time; they have different prefixes from other nouns of classes 1/2, but agreement with them is regular 1/2 agreement as shown here.

Table 1: Xhosa noun class markers and agreement

class	noun	gloss	X _{class.α} will fall	I saw <pronoun> class.α	adjectival concord
1	um-ntu	person	u-zo-wa 1SM-FUT-fall	Nd-a-m-bona 1SSM-PST-1OM-see	um-ntu om-khulu 1-person 1-big
2	aba-ntu	people	ba-zo-wa 2SM-FUT-fall	Nd-a-ba-bona 1SSM-PST-2OM-see	aba-ntu aba-khulu 2-people 2-big
1a	u-mama	mother	u-zo-wa 1SM-FUT-fall	Nd-a-m-bona 1SSM-PST-1OM-see	u-mama om-khulu 1-mother 1-big
2a	oo-mama	mothers	ba-zo-wa 2SM-FUT-fall	Nd-a-ba-bona 1SSM-PST-2OM-see	oo-mama aba-khulu 2-mothers 2-big
3	um-thi	tree	u-zo-wa 3SM-FUT-fall	Nd-a-wu-bona 1SSM-PST-3OM-see	um-thi om-khulu 3-tree 3-big
4	imi-thi	trees	i-zo-wa 4SM-FUT-fall	Nd-a-yi-bona 1SSM-PST-4OM-see	imi-thi emi-khulu 4-trees 4-big
5	ili-tye	stone	li-zo-wa 5SM-FUT-fall	Nd-a-li-bona 1SSM-PST-5OM-see	ili-tye eli-khulu 5-stone 5-big
6	ama-tye	stones	a-zo-wa 6SM-FUT-fall	Nd-a-wa-bona 1SSM-PST-6OM-see	ama-tye ama-khulu 6-stones 6-big
7	isi-tya	dish	si-zo-wa 7SM-FUT-fall	Nd-a-si-bona 1SSM-PST-7OM-see	isi-tya esi-khulu 7-dish 7-big
8	izi-tya	dishes	zi-zo-wa 8SM-FUT-fall	Nd-a-zi-bona 1SSM-PST-8OM-see	izi-tya ezi-khulu 8-stones 8-big
9	in-komo	cow	i-zo-wa 9SM-FUT-fall	Nd-a-yi-bona 1SSM-PST-9OM-see	in-komo en-kulu 9-cow 9-big
10	iin-komo	cows	zi-zo-wa 10SM-FUT-fall	Nd-a-zi-bona 1SSM-PST-10OM-see	iin-komo ezi-khulu 10-cows 10-big

2.2 Semantics of noun classes

2.2.1 Proto-Bantu

Strands of semantic unity have led to reconstructions in which at least some Proto-Bantu classes have regular meanings associated with them. Table 2 is fairly standard in this regard.

Table 2: A Proto-Bantu noun class reconstruction (Vossen & Dimmendaal 2020:151)

Singular (number) ⇄	Singular (form) ⇄	Plural (number) ⇄	Plural (form) ⇄	Semantics ⇄
1	*mù-	2	*βà-	humans
3	*mù-	4	*mì-	trees, plants
5	*lì-	6	*mà-	mixed/cl. 6 liquids
7	*kì-	8	*βì-	mixed
9	*nì-	10	*lì-nì-	animals, mixed
11	*lù-			mixed
12	*kà-	13	*tù-	augmentative, diminutive, etc.
14	*βù-			abstract
15	*kù-			infinitive
16	*pà-			location on
17	*kù-			location at
18	*mù-			location in
19	*pì-			diminutive

In Xhosa noun classes there are a few semantic regularities consistent with Table 2: only [human] nouns bear the **um-/aba-** prefixes of classes 1/2, and kinship terms are consistently found in subclass 1a/2a bearing its distinct prefixes **u-/oo-**. But these are one-way correlations: [human] nouns are scattered across the classes, and in addition to kinship terms, 1a/2a include all the letters of the alphabet and a variety of animals and inanimates. Other semantic associations reconstructed in Table 1 are weak or lost. Liquids, tree and plant terms, and animals names occur in all classes. Diminutive, augmentative, and locative classes 16 and 18 are absent in Xhosa; locative class 17 is prepositional (Buell 2007) and will not be relevant here.

The heterogeneity of Xhosa classes is a significant factor in the analysis. In what follows I provide an overview of the distribution of the major semantic fields across classes.

2.2.2 The heterogeneous contents of Xhosa classes

(11) lists some representative examples of the distribution of [human] nouns in Xhosa.

In classes 3/4 there are only a few and they are stigmatized, but in 1/2, 5/6, 7/8, and 9/10 there

are very many, without predictable semantic connotations.³

(11) Humans: classes 1/2

- | | | |
|--|--|--|
| a. um -ntwana/ aba -ntwana
1-child/2-child
'child/ren' | b. um -ntu/ aba -ntu
1-person/2-people
'person/people' | c. um -/ aba -hleli
1-editor/2-editor
'editor/s' |
|--|--|--|

Humans: classes 3/4 (stigmatized)

- | | | |
|---|---|--|
| d. um -gulukudu/ imi -gulukudu
3-gangster/4-gangster
'thug/s, gangster/s' | e. um -gewu/ imi -gewu
3-criminal/4-criminal
'criminal/s' | f. um -lwelwe/ imi -lwelwe
3-/4-disabled or poor
'disabled or poor person/s' |
|---|---|--|

Humans: classes 5/6

- | | | |
|---|--|---|
| g. i -qhawe/ ama -qhawe
5-hero/6-hero
'hero/es' | h. i -gqwetha/ ama -gqwetha
5-lawyer/6-lawyer
'lawyer/s' | i. i -sela/ ame -sela
5-thief/6-thief
'thief/thieves' |
|---|--|---|

Humans: classes 7/8

- | | | |
|---|--|---|
| j. isi -bonda/ izi -bonda
7-headman/8-headman
'headman/men' | k. isi -hlobo/ izi -hlobo
7-friend/8-friend
'friend/s' | l. is -anuse/ iz -anuse
7-diviner/8-diviner
'diviner/s' |
|---|--|---|

Humans: classes 9/10

- | | | |
|---|--|---|
| m. in -tombi/ iin -tombi
9-young.lady/10-young.lady
'young lady/ladies' | n. in -gcali/ iin -gcali
9-expert/10-expert
'expert/s' | o. im -bongi/ iim -bongi
9-poet/10-poet
'poet or praise singer/s' |
|---|--|---|

[Animal] nouns are also well-represented in every pair of noun classes as (12)

demonstrates. Classes 9/10 seem to include a plurality of them, however.

(12) Animals: classes 1a/2a

- | | | |
|---|--|---|
| a. u -nokala/ oo -nokala
1a-crab/2a-crab
'crab/s' | b. u -krebe/ oo -krebe
1a-shark/2a-shark
'shark/s' | c. u -dyakalashe/ oo -dyakalashe
1a-jackal/2a-jackal
'jackal/s' |
|---|--|---|

Animals: classes 3/4

- | | | |
|--|---|---|
| d. um -khombe/ imi -khombe
3-rhino/4-rhino
'rhino/s' | e. um -qhagi/ imi -qhagi
3-rooster/4-roosters
'rooster/s' | f. um -thwane/ imi -thwane
3-donkey/4-donkey
'donkey/s' |
|--|---|---|

Animals: classes 5/6

- | | | |
|--|---|--|
| g. i -cikilishe/ ama -cikilishe
5-lizard/6-lizard
'lizard/s' | h. i -hashe/ ama -hashe
5-horse/6-horse
'horse/s' | i. i -hobe/ ama -hobe
5-dove/6-dove
'dove/s' |
|--|---|--|

³ Mletshe (2019) illustrates some semantic regularities correlating with output classes of deverbal noun formation. As these are not characteristic of basic, underived nouns they are somewhat orthogonal to my concerns in this paper; space precludes full discussion but I suggest that semantic phenomena are properties of the particular nominalizers and nominalization processes involved – see Carstens (2008) for an approach.

Animals: classes 7/8

j. **isi-gcawu/izi-gcawu**
7-spider/8-spider
'spider/s'

k. **isi-khova/izi-khova**
7-owl/8-owl
'owl/s'

l. **isi-lwanyana/izi-lwanyana**
7-animal/8-animal
'animal/s'

Animals: classes 9/10 (plurality of animals; a fairly predictable mapping)

m. **in-ja/iin-ja**
9-dog/10-dog
'dog/s'

n. **in-dlovu/iin-dlovu**
9-elephant/10-elephant
'elephant/s'

o. **i-hagu/ii-hagu**
9-pig/10-pig
'pig/s'

[Inanimate] nouns are similarly scattered. For example, though many are found in 7/8, the generic term for 'thing(s)' is a class 9/10 noun.

(13) Inanimates: 1a/2a

a. **u-lolilwe/oo-lolilwe**
1a-train/2a-train
'train/s'

b. **u-matshini/oo-matshini**
1a-machine/2a-machine
'machine/s'

c. **u-L/oo-L**
1a-L/2a-L
'letter L/s'

Inanimates: 3/4

d. **um-bhobho/imi-bhobho**
3-pipe/4-pipe
'pipe/s'

e. **um-nqwazi /imi-nqwazi**
3-hat/4-hat
'hat/s'

f. **um-ba/imi-ba**
3-issue/4-issues
'issue/s'

Inanimates: 5/6

g. **i-qhosha/ama-qhosha**
5-button/6-6-button
'button/s'

h. **i-cepe/ama-cepe**
5-spoon/6-spoon
'spoon/s'

i. **i-gama/ama-gama**
5-word/6-word
'word/s or name/s'

Inanimates: 7/8

j. **isi-bane/izi-bane**
7-lamp/8-lamp
'lamp/s'

k. **isi-tya/izi-tya**
7-dish/8-dish
'dish/es'

l. **isi-Xhosa**
7-Xhosa
'Xhosa language'

Inanimates: 9/10

m. **in-to/izin-to**
9-thing/10-thing
'thing/s'

n. **in-cwadi/iin-cwadi**
9-book/10-books
'book/s'

o. **in-daba/iin-daba**
9-news/10-news
'news item/s'

The generic term for 'tree' is *umthi*, a noun of classes 3/4 as is consistent with the proto-Bantu reconstructions of these classes' contents, and many tree terms are of the form *umthi wa X* - 'tree of X', especially those that bear fruit. Specific names for species of trees are often of class 3/4 as well, but quite a lot fall outside it as exemplified below. The generic term for 'plant' is a deverbal noun of classes 7/8 (*isi-tyalo* < *uku-tyala* -- 'to plant'), and generic 'bush' is a noun

of 5/6. I have yet to find plant terms in classes 1a/2a.

(14) Plants and trees: 3/4

- | | | |
|---|--|--|
| a. um -thi/ imi -thi
3-tree/4-tree
'tree/s' | b. um -hlonyane/ imi -hlonyane
3-Wormwood/4-Wormwood
'Wormwood tree/s' | c. um -nquma/ imi -nquma
3-wild.olive/4-wild.olive
'wild olive tree/s' |
|---|--|--|

Plants and trees: 5/6

- | | | |
|--|---|---|
| d. i -khala
5-aloe
'aloe/s' | e. i -tyholo/ ama -tyholo
5-bush/6-bushes
'bush/es' | f. i -cuba
5-tobacco
'tobacco' |
|--|---|---|

Plants and trees: 7/8

- | | | |
|--|---|--|
| f. isi -thungwa/ izi -thungwa
7-sew.PASS/8sew.PASS
'Cape Quince' (plaited for use) | g. isi -tyalo/ izi -tyalo
7-plant.NOM/8-plant.NOM
'plant/s' | h. isi -Qwane/ izi -Qwane
7-Protea/8-Protea
'Protea' |
|--|---|--|

Plants and trees: 9/10

- | | | |
|--|--|---|
| i. in -tshilo/ iin -tshilo
9-TSHILO /10-TSHILO
'Hedge Caper' | j. in -yibiba/ iin -yiba
9-lily/10-lillies
'lily/lilies' | k. in -gcongolo/ iin -gcongolo
9-reed/10-reeds
'reed/reeds' |
|--|--|---|

(15) shows that liquid and mass nouns are quite scattered across the Xhosa noun classes.

(15) Liquids and masses: scattered

- | | | | | | |
|---|--|--|---|-------------------------------------|--|
| a. um -chamo
3-urine
'urine' | b. i -gazi
5-blood
'blood' | c. ama -nzi
6-water
'water' | d. isi -dudu
7-porridge
'porridge' | e. i -oili
9-oil
'oil' | f. u -bisi
11-milk
'milk' |
| g. i -tyuwa
9-salt
'salt' | h. um -gubo
3-flour
'flour' | i. ama -zimba
6-sorghum
'sorghum' | j. u -daka
11-mud
'mud' | | |

In some languages agreement with clauses reveals the default gender but this is not the case in Xhosa, where clauses and gerunds belong to a dedicated class 15 (see (16)).

- (16) Clauses class 15
- | | | |
|--|--|---|
| a. Ndi-ya- ku -cinga
1SSM-DISJ- 15om -think that
'I think that Sabelo left.' | ukuba u-hambile uSabelo.
1SM-left 1a-Sabelo | b. Uku-cula ku -mnandi.
15-sing 15sm -nice
'Singing is nice.' |
|--|--|---|

Agreement is strictly based on class, not semantic features (see §7 for further details):

- | | | |
|--|---|--|
| (17) a. um -fazi w -a-fika.
1-woman 1SM -PST-arrive
'The woman arrived.' | b. in -tombi y -a-fika.
9-girl 9SM -PST-arrive
'The girl arrived.' | c. i -gwetha l -a-hleka.
5-lawyer 5SM -PST-laugh
'The lawyer laughed.' |
| (18) a. u -lolilwe w -a-fika.
1a-train 1SM -PST-arrive
'The train arrived.' | b. i -posi y -a-fika.
9-mail 9SM -PST-arrive
'The mail arrived.' | c. ili -tya l -a-wa phantsi.
5-stone 5SM -PST-fell down
'The stone fell.' |

- (19) a. **aba**-fazi/**oo**-lolilwe **ba**-a-fika.
 2-women/2a-trains 2SM-PST-arrive
 'The women/trains arrived.'
- b. **iin**-tombi/**iin**-cwadi **z**-a-fika
 10-girls/10-letters 9SM-PST-arrive
 'The girls/letters arrived.'
- c. **ama**-gwetha/**ama**-tye a-a-wa.
 6-lawyers/6-stones 6SM-PST-fell
 'The lawyers/stones fell.'
- d. **imi**-thi/**imi**-gewu y-a-wa.
 4-trees/4-criminals 4SM-PST-fell
 'The trees/criminals fell down.'

3. Agreement with conjoined singulars

3.1 "Default"/semantic agr with mismatching [sg+sg]: classes 2 and 8

Given their highly varied contents, consistent semantic features seem not to be characteristic of Xhosa noun classes apart from a few limited strands of coherence; but as §1 pointed out, agreement with conjoined singulars reveals a conceptual division of [human] vs. others. We see this manifested in the Xhosa (20)a,b where singular nouns from different classes are conjoined. In (20)a where the conjoined nouns denote inanimates, agreement is class 8. For conjoined [human] nouns of mismatching classes agreement is class 2.

- (20) a. U-m-nqathe ne-qanda **zi**-se tafile-ni. [Xhosa; Mitchely 2015:115]
 3-carrot and.5-egg 8SM-LOC table-LOC
 'The carrot and the egg are on the table.'
- b. U-m-fundi ne-polisa **ba**-ya-baleka [Xhosa; Mitchley 2015:115]
 1-1-student and.5-policeman 2SM-DISJ-run
 'The student and the policeman are running.'

The pattern is common across Bantu languages -- agreement with mismatched conjoined nominals is quite generally taken from plural noun class 2 for [human] nouns and from class 8 for [inanimate], regardless of the conjuncts' intrinsic classes (Corbett 1983, 1991, 2006, Corbett & Mtenje 1987, De Vos & Mitchley 2012, Diercks et al 2015, Givon 1970, Marten 2000, 2005, Roberts & Wolontis 1974, Schadeberg 2001, Simango 2012, and Voeltz 1971, Mitchley 2015). (21) and (22) provide some examples from the literature (glosses adapted); (23) gives a baseline descriptive generalization.

- (21) a. Mbale na ka-temo **v**-a-sowa. [ciNsenga; Simango 2012: 178]
 9.plate and 12-axe 8SM-past-miss
 'The plate and the axe are missing.'

- b. Li-doda na m-simbi **w**-e-fwika mailo [ciNsenga; Simango 2012:179]
 5-man and 1-girl 2SM-PST-arrive yesterday
 'The man and the girl arrived yesterday.'
- (22) a. U-mu-bíirá ní-í-ri-gena m-**bí**-síir:e [Kuria; from Diercks et al 2015:31]
 3-3-ball and-5-5-stone FOC-8SM-disappear.PST
 'The ball and the stone disappeared.'
- b. Axi-kelema no buchara **vo** vulavula. [Xitsonga; Mitchley 2015:65]
 DEF-7-scoundrel and 9.butcher 2SM.PRES talk
 'The scoundrel and the butcher are talking.'
- (23) Default or semantic agreement rules for Bantu, commonly assumed:
 a. For conjoined mismatched [+human] nouns, agreement is class 2.
 b. For conjoined mismatched [-human] nouns, agreement is class 8.

Though I will use the term default agreement for convenience, there are clearly semantic underpinnings to this strategy as has been widely noted in the literature since, as we have seen, classes 1/2 are reconstructed as the canonical domain of [human] nouns and while the reconstruction in Table 1 characterizes the contents of classes 7/8 as mixed, these are often characterized as the canonical [inanimate] classes (Denny & Creider 1986 among others).

3.2 Cases where a default strategy is puzzling

3.2.1 Singular inanimates [3+3], [5+5] pair with class 8 *zi*-

As pointed out in Taraldsen et al (2018) and mentioned in §1, some pairs of conjoined singulars drawn from a single class – in particular the combinations [3+3] and [5+5] – also take these same *ba*- and *zi*- defaults in Xhosa. (24)a and (25)a show this for conjoined [inanimate] nouns: agreement with them is class 8 *zi*-, though as (24)b and (25)b demonstrate, the same nouns' plurals belong to class 4 and 6 respectively, and pair with the agreement markers *i*- and *a*-.

- (24) a. Um-nqwazi nom-pu √**zi**- /X**i**- se tafile-ni. [3+3=8; ≠4]
 3-hat and.3-gun 8SM/ 4SM-are table-LOC
 'A hat and a gun are on the table.'
- b. Imi-nqwazi X**zi** /√**i**-se tafile-ni. [plural of N_{cl.3} is cl 4: *i*-agr]
 4-hats 8SM/ 4SM-be table-LOC
 'The hats are on the table.'
- (25) a. Ili-tye ne-qanda √**zi**-/X**a**-nyamalele. [5+5=8, ≠ 6]
 5-stone and.5-egg 8SM/6SM-disappeared
 'The stone and the egg disappeared.'

- b. Ama-tye Xzi /√a-nyamalele.
 6-stones 8SM/ 6SM-disappeared
 'The stones disappeared.'

[plural of N_{cl.5} is cl 6: α-agr]

3.2.2 Conjoined humans of [3+3], [5+5] pair with class 2 *ba-*

Similarly, conjoined [human] nouns of these classes pair with the class 2 prefix *ba-* rather than with the agreement markers accompanying their plurals ((26)a,b and (27)a,b).

- (26) a. Um-gewu nom-gulukudu √**ba** /X*i*-sebenza ndawonye. [3+3=2, ≠4]
 3-criminal and.3-gangster 2SM/ 4SM-work together
 'A criminal and a gangster are working together.'
- b. I-mi-gewu X**ba**/√*i*-sebenza ndawonye.
 4-criminals 2SM/ 4SM-work together
 'The criminals work together.'
- (27) a. I-gqirha ne-gosa √**ba** /X*a*-sebenza ndawonye. [5+5=2, ≠6]
 5-healer and.5-officer 2SM/ 6SM-work together
 'The healer and the officer are working together.'
- b. A-ma-gqirha X**ba**/√*a*-sebenza ndawonye.
 6-healers 2SM/ 6SM-work together
 'The healers are working together.'

Taraldsen et al (2018) conclude from this that Bantu singular/plural pairings do not share gender features. Each singular and each plural class is a distinct gender.

3.3 Analyses of parallel facts in canonical gender systems.

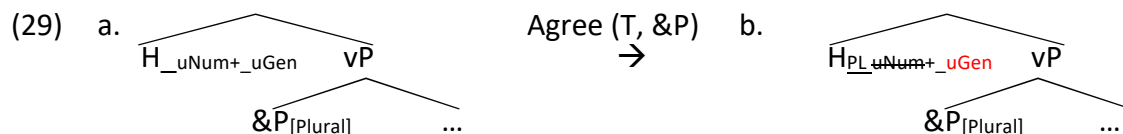
Contrary to Taraldsen et al's claim, the patterns of default agreement in (24)-(27) do threaten the gender analysis of pairs of classes owing to well-known parallels in languages with more canonical three-gender systems. As noted in §1, neuter singulars in BCS do not pair with plural neuter agreement but rather with default masculine as demonstrated in (28) below:

- (28) *Jedno tele i jedno pašče su juče prodana. [Bošković 2009]
 one calf.neut and one dog.neut are yesterday sold.pl.neut
 Intended: a calf and a dog were sold yesterday.'

Marušić et al (2007) and Bošković (2009) propose an account: Conjunct Phrase (&P) has number features only, and it's a closer goal to a clause-level agreement probe than are &P's contents. It therefore intervenes to block access to the conjuncts' features. A uPhi probe on a

head H obtains from &P a plural value alone, leaving the gender probe unvalued.

Default/semantic resolution rules follow whether the genders of conjuncts mismatch or match.



3.4 Where regular plural agreement obtains

3.4.1 Conjoined singulars [1+1], [7+7], [9+9] pair with expected plural agreement

There is a fatal flaw in this approach however: (29) leads to an expectation that agreement will

never match the gender of conjoined singulars. Yet as I mentioned in the introduction,

agreement *does* match, in certain genders. In Xhosa, conjoined singular nouns of class 1 take

class 2 *ba*-AGR (see (30) and (31)). This is true whether or not they are [human], arguing that the

pattern cannot be attributed to their intrinsic semantics triggering the default rule in (23)a.

(30) Um-mi nom-ongameli **ba**-ya-ncokola. [1+1=2]
 1-citizen and.1-president 2SM-DISJ-chat
 'The citizen and the president are chatting.'

(31) a. U-L no-M **ba**-/*zi-se tafile-ni. [1+1=2]
 1a-L and.1a-M 2SM-/*8SM-LOC table-LOC
 'The L and the M are on the table.'

b. U-loliwe kunye no-matshini **ba**-/*zi-ya-hamba. [1+1=2]
 1a-train and and.1a-machine 2SM-/*8SM-DISJ-move
 'The train and the machine are moving.'

Similarly, conjoined singulars of class 7 take class 8 *zi*-agr whatever they denote; thus

conjoined [human] nouns in class 7 do not pair with the class 2 *ba*- that serves as default for

conjunctions of [human] nouns from classes [3+3] and [5+5].

(32) Isi-bane nesi-tya **zi**-nyamalele. [7+7=8]
 7-lamp and.7-dish 8SM-disappeared
 'The lamp and the dish have disappeared.'

(33) Is-anuse nes-azi **zi**-ya-sebenza.
 7-diviner and.7-scientist 8SM-DISJ-work
 'The diviner and the scientist are working.'

Conjoined singulars of class 9 also take *zi*-AGR, even if [human]:

(34) In-dlovu (kunye) nen-gwe zi-ya-lwa. [9+9=10]
 9-elephant and and.9-leopard 8SM-DISJ-fight.
 'The elephant and the leopard are fighting.'

(35) In-dadi nen-tlebi zi-ya-cula.
 9-swimmer and.9-gossip 10SM-DISJ-sing
 'The swimmer and the gossip are singing.'

Class 8 and 10 agreement morphology is syncretic; *zi-* is ambiguous between the two classes. I analyze the *zi-* accompanying conjoined class 9 singulars as class 10 rather than default class 8 since it pairs with conjunctions of [human] nouns like those in (35), for which the default is not *zi-* but rather *ba-*. In other words this is formal grammatical agreement, like class 2 agreement with the same range of [human], [inanimate], and [animal] nouns.

3.4.2 More parallels in Slovenian and BCS

This pattern too has parallels outside of Bantu: as we have seen, though BCS [&P neut+neut] take default masculine, [&P fem+fem] and [&P masc+masc] can take matching agr ((3) from Murphy & Puskar 2018; (4) from who? glosses adapted).

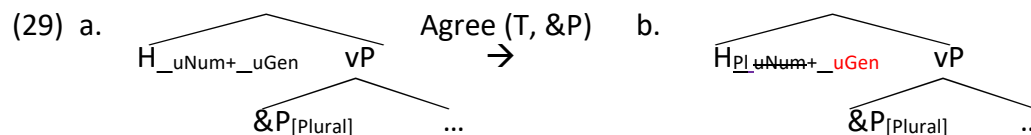
(36) Ovaj covek i onaj decak su stigli-i. [BCS: Despic 2017: 291]
 this man.MASC and that boy.MASC are arrived. MASC.PL
 'This man and that boy arrived.'

[&P MASC +MASC = MASC.PL]

(3) [&P Zavesa i biljka] su ukrašavale prozor.
 curtain.F.SG and plant.F.SG are decorate.PRT.F.PL window
 'A curtain and a plant decorate the window.'

[&P FEM +FEM = FEM.PL]

The facts are at odds with proposal (29) (repeated below).



Some recent works address the contrast between BCS neuter and the other genders: Despić (2016), Nevins (2018), Tsimpli & Hulk (2013), Adamson & Anagnostopolou (2024) propose that neuter is the absence of gender or negative values [-FEM, -MASC]. Such approaches seem reasonable for languages in which there is a single outlier gender. But the ideas do not extend

well to Xhosa, where there are two outliers, and as we will see in §7, Shona has six.

Bošković (2009) suggests that feminine but not neuter is reflected in agreement with conjoined singulars because feminine is semantically grounded and therefore percolates to &P, though why this should be so in cases like (3) he leaves open. I'll build on his proposal an analysis that takes the patterns to be semantically based, even in such cases.

4. Analysis

4.1 Why 1/2, 7/8, and 9/10 are different from 3/4, 5/6

I summarize in (37) the patterns to be captured:

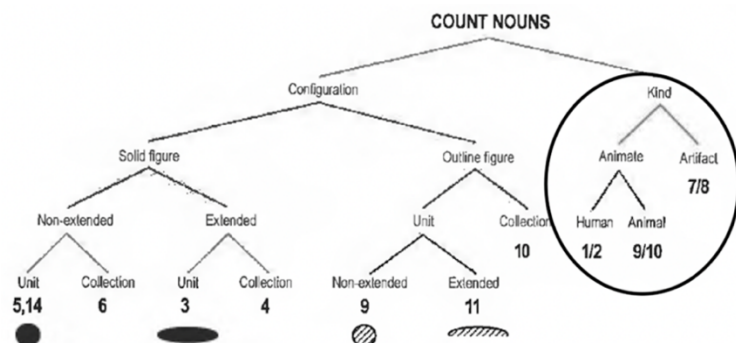
- (37) a. Conjoined singulars [1+1] [7+7], [9+9] control matching plural agreement.
b. [3+3] and [5+5] take default agreement.
c. Classes 2 and 8 (plurals of 1 and 7) provide default agreement for [3+3], [5+5] and for mismatching [X_{sing} & Y_{sing}].

Thus two of the genders exhibiting grammatical agreement with conjoined, gender-matched singulars are also the sources of default plural agreement -- classes 1/2, 7/8 -- and the defaults correlate with semantic associations of classes 1/2 to [human], and 7/8 to [inanimate]. By logical extension, the pattern [9+9=10] suggests that 9/10 have a semantic association as well; I assume that this is [animal] based on the reconstructed contents of this class as well as its inclusion of a large number of animal names synchronically. This makes 1/2, 7/8, and 9/10 the natural class of genders with semantic "flavors". Why though are 3/4 not part of this grouping, given their standard reconstruction as the classes of trees and plants?

It turns out that there is an alternative reconstruction of noun class meanings with precisely this exclusion. Denny & Creider (1986) propose that only 1/2, 7/8, and 9/10 have concrete semantic associations; 3/4, 5/6 and others are characterized only by abstract shape tendencies. I reproduce their schematic for this in Figure 1 in which I circle the 'kind' node and its

dependents. Denny & Creider argue that the number of reconstructed tree and plant names in proto-Bantu classes 3/4 is statistically insufficient to justify their inclusion beneath this node.⁴

Figure 1: Proto-Bantu noun class semantics in Denny & Creider (1986)

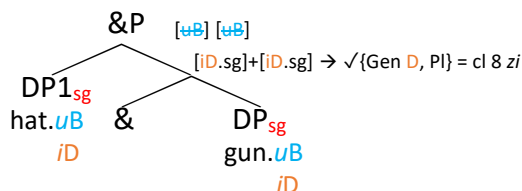


I analyze 3/4 and 5/6 as *ugenders* as they do not figure into resolved agreement which, following Adamson & Anagnostopoulou (2024), is based on interpretable features alone as will be detailed in §4.3. (38) illustrates the structure of a class 3/4 noun, and (40) previews how default agreement in (39) is derived via deletion of *ugender* in the resolution process.

(38) um-qwazi/imi-qwazi [n_B [n_D √QWAZI]] *un* of 3/4 stacks above 7/8
 3-hat/4-hat
 'hat/s'

(39) a. um-nqwazi nom-pu √zi- /X_i- se tatile-ni. [3+3=8; ≠4]
 3-hat and.3-gun 8_{SM}/ 4_{SM}-are table-LOC
 'A hat and a gun are on the table.'

(40) singular conjuncts trigger the resolution process



class 8 agr with conjoined inanimates [3+3]

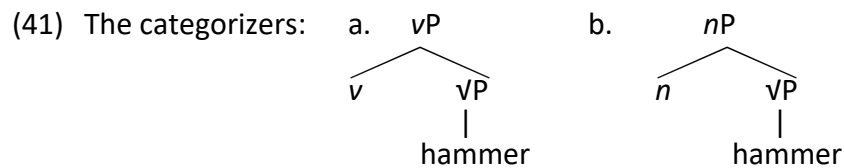
§7 will explore the properties of classes 3/4 and 5/6 in greater detail, once the analysis has been presented in full. It has been claimed that the unagreeability of conjoined neuter singulars

⁴ Xhosa botany: Bhatt (2013) lists 30 plants and trees of which 8 are in classes 3/4 = 26%; Bhatt (2014) lists 11/29 in class 3/4 -- 61%. In Wehmeyer & Rose (1983) plants and trees in classes 3/4 number 3/18 or 17%.

is related to the fact that neuter nouns are mass rather than count, hence largely unindividuated and unpluralizable (Arsenijevic 2017, 2021; Adamson & Anagnostopoulou 2024). In principle it might be that the shape associations proposed by Denny & Creider make 3/4 and 5/6 nouns similarly incompatible with resolution, but we will see that there are no syntactic indicators of defectiveness for nouns of Xhosa classes 3/4 and 5/6, unlike what Arsenijevic demonstrates for neuters; absent other diagnostics, I shelve the approach as unfalsifiable.

4.2 Gender, *n*, and interpretability

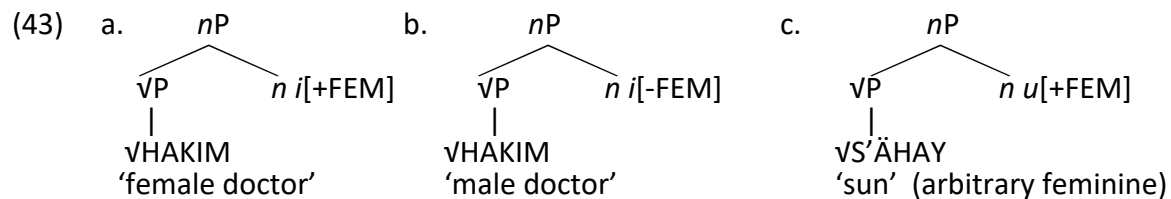
Following Kramer (2015), gender is a feature of the categorizer *n* as shown in (41):



Kramer proposes that genders may have *i*- and *u*flavors. Amharic has two genders: feminine and masculine, with *i*flavors [+FEM] and [-FEM] for female- and male-denoting nouns respectively. Nouns that are arbitrarily feminine have *u*[+FEM] gender, and the remainder are masculine, by default. I summarize in (42) and illustrate in (43) with two Amharic nouns naming entities with natural gender, and the arbitrarily feminine noun ‘sun’:

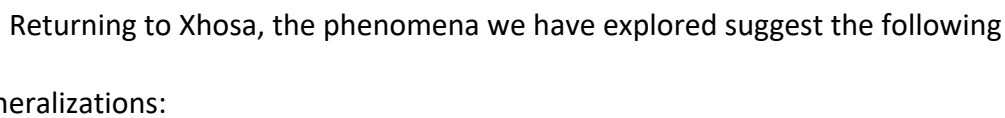
(42) Amharic: two genders. Types of *n*:

- a. *n i* [+FEM] Female natural gender
- b. *n i* [-FEM] Male natural gender
- c. *n u* [+FEM] Feminine arbitrary gender
- d. *n* No natural gender = “plain” *n* (grammatically masculine, by default)



I adopt this approach (with a modification below), and a second tool from Kramer: gender

(44) a. ínan 'son, boy (m.)' [Somali]
b. inammó 'sons, boys (f.)'



- In Kramer's account, licensing conditions ensure that combinations of gender+root are compatible. For any u flavor, the relevant licensing condition applies at PF. This is consistent with the evidence that Xhosa's wholly u genders 3/4 and 5/6 never factor into the calculus of feature-resolution on &P, giving rise to "default" agreement. We need only assume that they are deleted from the syntactic representation in the resolution process, though their PF features remain.

⁵ See Carstens (2019) for a first sketch of this approach to Xhosa default/semantic agreement; Fuchs & van der Wal (2022) for analysis of Bantu diminutives and locatives via *ns* and *nP* stacking.

the success of appropriate plural agreement with all conjunctions [1+1], [7+7] and [9+9], I depart from this in proposing instead that any [gender+root] combination involving a gender that has an *i*flavor is subject to LF licensing conditions. This seems a natural state of affairs to ensure that *u*flavors pair with all and only roots whose membership in the relevant gender is arbitrary, and *i*flavors pair with roots of compatible conceptual content.

Recall the system of genders proposed in Carstens (1991, 2008) sketched in (8) and reproduced below.

(8) Bantu Genders (Carstens 1991)

Gender A: stems of Classes 1/2
Gender B: stems of Classes 3/4
Gender C: stems of Classes 5/6

Gender D: stems of Classes 7/8
Gender E: stems of Classes 9/10
Etc.

Genders A and D, underlying classes 1/2 and classes 7/8, have flavors *i*[human] and *i*[inanimate] respectively, accounting for two things: their survival in agreement with matched singular conjuncts [1+1] and [7+7], and their use as defaults for conjunctions of singular human and inanimate-denoting nouns. I have posited an *i*[animal] flavor for Gender E of classes 9/10 to account for class 10 agreement with conjoined singulars of class 9. Genders B and C, underlying 3/4 and 5/6 respectively, lack interpretable flavors as shown in (47). For expository ease I identify "flavors" other than the canonical semantic content of a gender as *arb* to distinguish them from the wholly *u*genders 3/4 and 5/6, though 3/4 and 5/6 are in fact always arbitrary, and the flavors A2, D2, and E2 are uninterpretable though this is opaque in narrow syntax.

(47) *ns* for Xhosa, reflecting conjunct agreement facts

Classes 1/2 = Gender A	<i>n</i> _{A1}	<i>i</i> [human]
	<i>n</i> _{A2}	<i>arb</i>
Classes 3/4 = Gender B	<i>n</i> _B	uninterpretable, for all members
Classes 5/6 = Gender C	<i>n</i> _C	uninterpretable, for all members
Classes 7/8 = Gender D	<i>n</i> _{D1}	<i>i</i> [artifact]
	<i>n</i> _{D2}	<i>arb</i>

Classes 9/10 = Gender E n_{E1} *i* [animal]
 n_{E2} *arb*

Comparing the synchronic contents of Xhosa classes to the Proto-Bantu reconstructions in Tables 1 and 2, it seems that large numbers of [+human] nouns drifted from Proto-Bantu classes 1/2 into other classes. Let's assume their new genders stack above an *icore* of Gender A.

(48) Sample structures of [human] nouns: a core of 1/2 = gender A

- a. *um-ntwana/aba-ntwana* [n_{A1} $\vee$ MNTWANA] n_A for [human]
 1-child/2-children
 'child/ren'
- b. *um-gewu/imi-gewu* [n_B [n_{A1} $\vee$ GEWU]] un_B of 3/4 stacks above 1/2 n_A
 3-criminal/4-criminal
 'criminal/s'
- c. *i-butho/ama-butho* [n_C [n_{A1} $\vee$ BUTHO]] un_C of 5/6 stacks above 1/2 n_{A1}
 5-warrior/6-warrior
 'warrior/s'
- d. *isi-hlobo/izi-hlobo* [n_{D2} [n_{A1} $\vee$ HLOBO]] n_{D2} of 7/8 stacks above 1/2 n_{A1}
 7-friend/8-friend
 'friend/s'
- e. *in-tombi/iin-tombi* [n_{E2} [n_{A1} $\vee$ TOMBI]] n_{E2} of 9/10 stacks above 1/2 n_{A1}
 9-young.lady/10-young.lady
 'young lady/ladies'

Similarly, I propose that [inanimate] nouns have class 7/8 cores. If their synchronic class membership is different, a higher n introducing an outer gender is layered above this:

(49) Structures of [inanimate] nouns: a core of 7/8 = gender D

- a. *isi-bane/izi-bane* [n_{D1} $\vee$ BANE] n_{D1} for [artifact]
 7-lamp/8-lamp
 'lamp/s'
- b. *u-matshini/oo-matshini* [n_{A2} [n_D $\vee$ MATSHINI]] n_{A2} of 1/2 above n_{D1} of 7/8
 1a-machine/2a-machine
 'machine/s'
- c. *um-qwazi/imi-qwazi* [n_B [n_D $\vee$ QWAZI]] un of 3/4 stacks above n_{D1} 7/8
 3-hat/4-hat
 'hat/s'
- d. *ili-tye/ama-tye* [n_C [n_D $\vee$ TYE]] un of 5/6 stacks above n_{D1} of 7/8
 5-stone/6-stone
 'stone/s'

[Animal] nouns outside of classes 9/10 have n_P layers above a gender E core as in (50).

(50) Structure of [animal] nouns: an inner layer of 9/10 = gender E

a. in -dlovu/ iin -dlovu 9-elephant/10-elephant 'elephant/s'	[n_{E1} VDLOVU]	n _{E1} for [animal]
b. u -nokala/ oo -nokala 1a-crab/2a-crab 'crab/s'	[n_{A2} [n_E VNOKALA]]	n _{A2} of 1/2 above n _{E1} of 9/10
c. um -vundla/ imi -vundla 3-rabbit/4-rabbit 'rabbit/s'	[n_B [n_E VQAZI]]	un of 3/4 above n _{E1} of 9/10
d. isi -khova/ izi -khova 7-owl/8-owl 'owl/s'	[n_D [n_E VQAZI]]	n _{D2} of 7/8 above n _{E1} of 9/10

5 Deriving the patterns

5.1 Conjoined singulars

I turn to the derivation of agreement with &P. In a study of Chichewa gender, Corbett & Mtenje (1987) propose that conjunctions of singulars and conjunctions with gender mismatch are both triggers for gender resolution. Interestingly, resolution for number forces resolution for gender. If conjuncts are plural and match in gender, resolution does not apply.

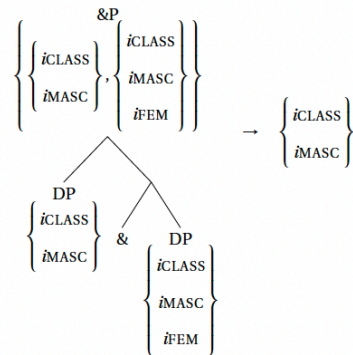
The correct generalization appears to be that gender resolution can be triggered in two ways. First it must operate if number resolution operates. There is a very general principle that if one resolution rule operates, all must operate if possible. This solution requires a stipulation in the number resolution rule that number resolution can operate only provided there is at least one singular conjunct. The second possible trigger for gender resolution is the presence of different genders in the subject....It followed as a logical consequence of this analysis that examples with conjoined masculine or conjoined feminine plurals ...do not undergo resolution... (Corbett & Mtenje 1987: 24)

That singular but not plural number triggers resolution in Xhosa is confirmed by comparison of gender-matching singulars and plurals of 3/4 and 5/6 (see (51), (52)).

- (51) a. Um-nqwazi nom-pu **zi**-se tafile-ni. b. I-gqirha ne-gosa **ba**-ya-sebenza.
 3-hat and.3-gun **8SM**-are table-LOC 5-healer and.5-officer **2SM**-DISJ-work
 'A hat and a gun are on the table.' 'The healer and the officer are working.'
- (52) a. Imi-pu nemi-nqwazi i-se tafile-ni.
 4-guns and.4-hats 4SM-are table-LOC
 'The guns and the hats are in the room.'

(55) Deriving masculine agreement in (121a):

Mismatch:



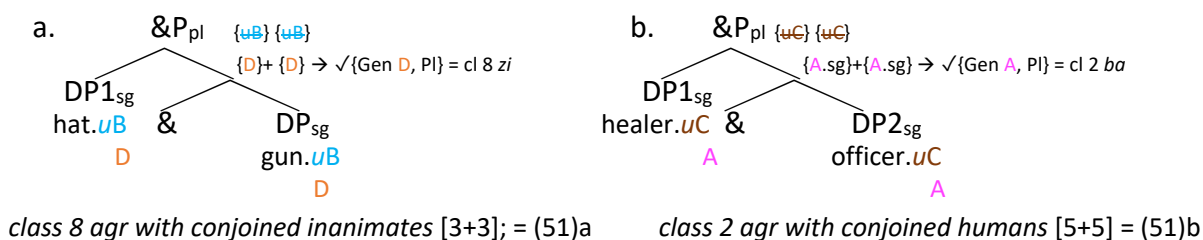
We do not find anything quite like these relationships among features in Bantu languages.

What I've analyzed as outer gender shells are not featurally more specific than the [human], [inanimate], [animal] genders that serve as defaults when nouns classes 3/4 or 5/6 are conjoined. Nor is there evidence of "nesting" dependency relations among these three semantic notions, which function as co-equals. In our culture, humans are views as a species of animal, but whether this view informs other languages is unclear.

The approach that I propose here in terms of Bantu gender-stacking is accordingly a bit different. But the mechanics of percolation and resolution apply very well.

I propose that default 2 and 8 agreement surface with conjoined singulars [3+3] and [5+5] because the wholly *ugenders* B and C are ignored in the calculus of resolution, as (56) illustrates. They delete, exposing the inner gender D core of [inanimate] nouns in (56)a = (51)a, and the gender A cores of [human] nouns in (56)b = (51)b. In each instance the gender of the inner core becomes the shared value on &P.

(56) Number and gender resolution triggered by [sing]+[sing].



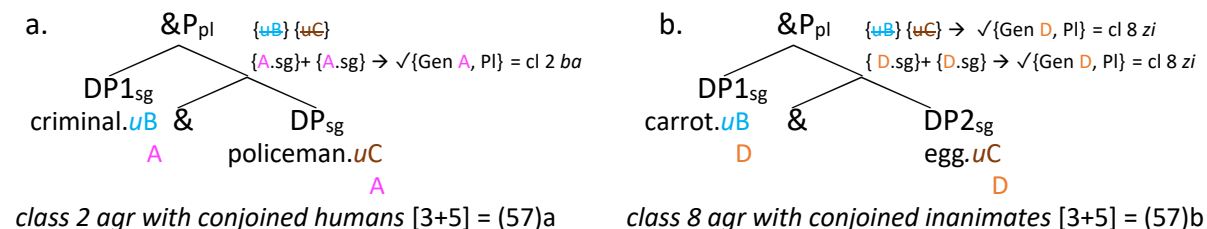
(57)a,b are coordinations of nominals that have mismatching *ugenders* as well as singular

number. Underlying these genders in each instance there are shared cores of gender A or D as

shown in (58)a,b. These control agreement once exposed by deletion of B and C.

- (57) a. Um-gewu ne-polisa **ba**-sebenza ndawonye. [Xhosa]
 3-criminal and.5-policeman 2SM-pres-work together
 'The criminal and the policeman are working together.'
- b. Um-nqathe ne-qanda **zi**-se tafile-ni. [Xhosa; Mitchely 2015:115]
 3-carrot and.5-egg 8SM-loc table-loc
 'The carrot and the egg are on the table.'

(58) Clashing genders, as well as singular number → deletion of *ugenders*



In cases like (59) and (60), matching arbitrary genders stack above *i*-gender cores. Recall my

proposal that their deletion is not possible before inspection by LF licensing conditions. Prior to

this point they have the same status as interpretable genders and are factored into resolution.

As these genders match and are highest, they "win" (Kramer 2015) (I indicate only the winner's

presence on &P for visual simplicity, though percolation of the lower gender features is

consistent with (56) and (58)).

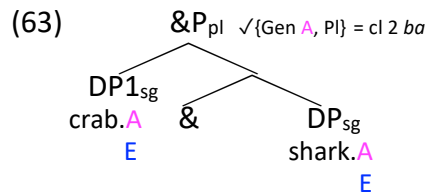
- (59) U-loliwe kunye no-matshini ba- /*zi-ya-hamba. [1a+1a=2]
 1a-train and and.1a-machine 2SM-/*8SM-DISJ-move
 'The train and the machine are moving.'

- (60) In-tombazana nen-kwenkwe zi- /*ba-ya-sebenza. [9+9=10]
 9-girl and.9-boy 10SM/*2SM-DISJ-work
 'The girl and the boy are working.'



Conjunctions of [animal] nouns follow the same patterns. If both are drawn from gender A, they pair with class 2 *ba*-agreement of gender A.

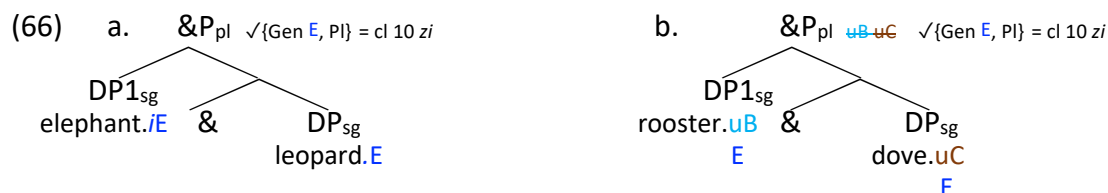
- (62) U-nokala no-krebe ba-ya-lwa
 1a-crab and.1a-shark 2SM-DISJ-fight
 'The crab and the shark are fighting.'



[Animal] nouns in class 9, like those of (64), have simple structures and when conjoined we have seen they take class 10 *zi*-agreement; this is derived as shown in (66)a. (66)b illustrates how gender mismatched [animal] conjuncts of the *u*genders such as those in (65) also combine with class 10 *zi*-agreement, based on their gender E cores.

- (64) In-dlovu (kanye) nen-gwe zi-ya-lwa. [9+9=10]
 9-elephant and and.9-leopard 10SM-DISJ-fight.
 'The elephant and the leopard are fighting.'

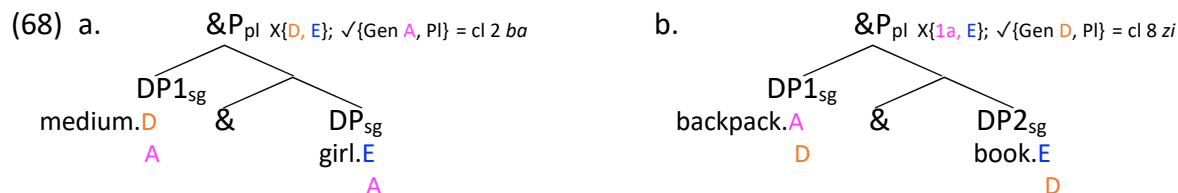
- (65) Um-qhagi ne-hobe zi-haphazela kanye.
 3-rooster and.5-dove 10SM-fly together
 'The rooster and the dove are flying together.'



In the conjunctions below, the outer genders have *i*flavors but they mismatch. Agreement is determined in the resolution process by finding their common denominators, the [human] and

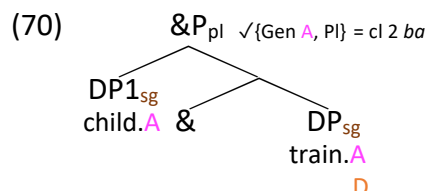
[inanimate] gender *icores*. The use of default class 2 *ba* in (67)a is consistent with the hypothesis that 7/8 and 9/10 are distinct genders even though class 8 and 10 agreements are identical.

- (67) a. Is-anuse nen-tombi ba-ya-sebenza. b. U-bhaka nen-cwadi zi-ngaphandle.
 7-medium and.9-girl 2SM-DISJ-work 1a-backpack and.9-book 8SM-be.outside
 'The medium and the girl are working.' 'The backpack and the book are outside.'



In (69), the two conjuncts share an outer gender which has an *iflavor*, though their inner cores differ. Agreement in the shared outer gender succeeds as illustrated in (70).

- (69) Um-twana no-lolilwe ba-ya-hamba
 1-child and.1a-train 2SM-DISJ-move
 'The child and the train are moving.'



Agreement with conjunctions of humans and inanimates is ineffable, if they are drawn from different genders. No choice of resolved agreement works in the following; a paraphrase like 'The girl collided with the train' is preferred.

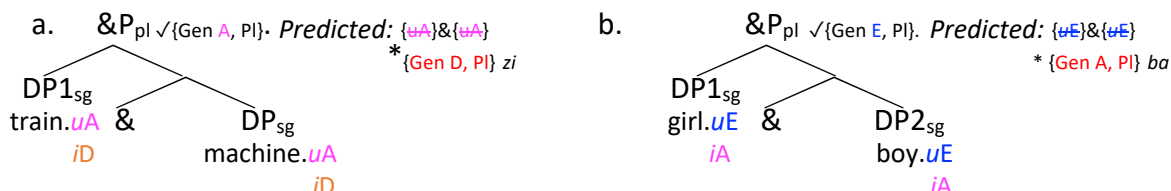
- (71) *In-tombi no-lolilwe zi/ba-gilana.
 9-girl and.1a-train 8SM-/2SM-collide
 'The girl and the train collided.'

The case is particularly interesting in that class 2 agreement is impossible here though each conjunct has either inner or outer gender A (see (72)). This argues that the higher genders are assessed as a set, and as they do not match, the lower genders are considered next. As they also mismatch, conversion fails.

(59) U-loliwe kunye no-matshini ba- /*zi-ya-hamba. [1a+1a=2]
 1a-train and and.1a-machine 2SM-/*8SM-DISJ-move
 'The train and the machine are moving.'

(60) In-tombazana nen-kwenkwe zi- /*ba-ya-sebenza. [9+9=10]
 9-girl and. 9-boy 10SM-/*2SM DISJ-work
 'The girl and the boy are working.'

(74) Predictions if *u*- and *i*-flavors of the same genders were distinguished in the morphosyntax



(75) a. Um-gewu nom-gulukudu ✓ba /Xi-sebenza ndawonye. [3+3=2, ≠4]
 3-criminal and.3-gangster 2SM/ 4SM-work together
 'A criminal and a gangster are working together.'

(76) a. I-gqirha ne-gosa ✓ba /Xa-sebenza ndawonye. [5+5=2, ≠6]
 5-healer and.5-officer 2SM/ 6SM-work together
 'The healer and the officer are working together.'

I conclude that arbitrary members of a gender do not pattern as this hypothesis would predict. Instead, all "flavors" of a gender pattern alike with respect to conversion and resolution. This motivates my proposal that if a gender has any interpretable content, both its *i*- and *u*-flavors are retained throughout the syntactic derivation.

6. Agreement with conjoined plurals: patterns and implications

6.1 Overview

In this section we will see evidence that resolution is never triggered by gender mismatch, contra Corbett & Mtenje (1987): in Xhosa, resolution does not apply to determine agreement with conjoined plurals at all. Not only does default agreement fail to appear in circumstances paralleling those where it is required with conjoined singulars; there is also variability in agreement with plurals whose genders mismatch to a degree not found with conjoined singulars. Both factors point towards a very different derivation without the resolution process. Despite this, agreement largely avoids any plural conjunct of noun class 4 or 6 in favor of those

drawn from class 2, 8, or 10. This parallels the inability of the corresponding singulars 3 and 5 to value agreement when conjoined, strongly supporting the analysis of 3/4 and 5/6 as connected -- as singulars and plurals of the same genders, contra Taraldsen et al (2018). The patterns that emerge are as follows, where (77)(i) bleeds (ii).

- (77) Hierarchies of preference in agreement with gender-mismatched conjoined plurals
 (i) Genders with *i*content > default > wholly *u*genders
 (ii) FCA > LCA

To capture these patterns I propose a Multiple Agree derivation in which the features of both conjuncts copy to the probe and are reduced to a single set for exponence, guided by these preferences. Multiple Agree is blocked if the conjuncts are singular because, as proposed by Marušič et al (2007) and Bošković (2009) and mentioned in §3.3, to interact with the conjuncts the probe must also target &P, and &P is plural. The conjuncts must be compatible with &P's plurality for Multiple Agree to succeed.⁶

6.2 When the genders match

As noted in §5.1, all gender-matching [plural+plural] conjunctions pair with expected grammatical agreement and not with defaults. We see this for classes 2, 8, and 10 in (78) where the pattern is not surprising, since singulars 1, 7, and 9 also pair with their expected plurals. But (79) and (80) show that the pattern holds even for [4+4] and [6+6]. Compare these examples with conjoined singulars of the same genders, i.e. [3+3] and [5+5] in (75) and (76) above; the conjoined singulars of these two genders pair with default *zi-* or *ba-*, depending on humanness.⁷

⁶ Taraldsen et al (2018) mention a small number of speakers offering agreement with a singular conjunct in lieu of resolved agreement (of approx. 20 speakers on each experiment, 1 speaker on one data set, 2 on another). Of the 8 speakers in my study none did so, but I have done some (remote) work since then with a speaker who sometimes offers FCA with a singular. This suggests that &P is not always plural for all speakers; how this impacts the overall grammar of agreement with conjuncts remains for future research.

⁷ This difference between singulars and plurals seems logically reducible to the fact that singular is additive, while plural is cumulative. Combining singulars returns something different -- a plural -- but combining of

- (78) a. Aba-ntwana noo-lolilwe ba-ya-hamba.
 2-children and.2a-train 2SM-DISJ-move
 'The children and the trains are moving.'
- b. Izi-tyebi nezi-bhanxa zi-ya-funda ndawonye. [Mitchley 2015: 116, adapted]
 8-rich.persons and.8-fools 8SM-DISJ-study together
 'The rich people and the fools are studying together.'
- c. lin-tombazana neen-kwenkwe zi- /*ba-ya-sebenza. [9+9=10]
 10-girls and.10-boys 10SM/*2SM-DISJ-work
 'The girl and the boys are working.'
- (79) a. Imi-pu nemi-bhobho i-se rumi-ni.
 4-guns and.4-pipes 4SM-LOC room-LOC
 'The guns and the pipes are in the room.'
- b. Ama-sele nama-dada a-ya-qubha.
 6-frogs and. 6-ducks 6SM-DISJ-swim
 'The frogs and the ducks are swimming.'
- (80) a. Ama-polisa nama-gqwetha a-ya-sebenza ndawoye.
 6-policemen and.6-lawyers 6SM-DISJ-work together
 'The policemen and the lawyers are working together.'
- b. Imi-gewu nemi-gulukudu i-se rumi-ni.
 4-criminals and.4-gangsters 4SM-LOC room-LOC
 'The criminals and the gangsters are in the room.'

The examples show us that the "problem" genders *can* factor into agreement with conjoined DPs just in case they belong to gender-matching plurals. The set of facts is surface compatible with Corbett & Mtenje's (1987) proposal that only singular number *or* mismatches in gender trigger resolution, since in conjunctions of [4+4] and [6+6], neither occur. But the next subsection presents reason to revise this.

6.3 When genders of plurals mismatch

6.3.1 FCA with 2, 8, 10

In agreement with gender-mismatched [pl+pl] combinations, first conjunct agreement (FCA) was preferred by a majority of the speakers in my study. (81) shows that this preference was absolute when the first conjunct was a human-denoting plural of class 2. When the first

plurals yields plurals; a calculation or transformation is involved in the first instance, but not in the second. I leave it to future research to explore this connection.

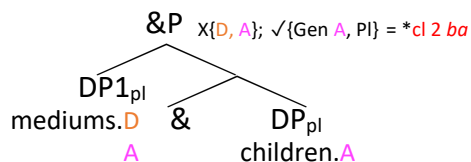
conjunct was instead class 8 or 10, the preference was not absolute but nonetheless strong.

Even when the second conjunct was class 2 as in (82)a,b, five and six out of eight speakers selected FCA.

- (81) Aba-ntwana neen-tombi ba-ya-cula. [2+10: FCA chosen by 8 out of 8 speakers]
 2-children and.10-girls 2SM-DISJ-sing-FV
 'The children and the girls are singing.'
- (82) a. lin-tombi naba-ntwana zi/ba-ya-cul-a. [10+2: FCA>LCA 5:3]
 10-girls and.2-children 10SM/2SM-DISJ-sing-FV
 'The young ladies and the children are singing.'
- b. Iza-nuse naba-ntwana zi-ya-cul-a. [FCA>LCA 6:2]
 8-mediums and.2-children 8SM-DISJ-sing-FV
 'The young ladies and the children are singing.'
- c. Izi-tya nemi-nqathe zi-se tatile-ni. [FCA>LCA 6:2]
 8-plates and.4-carrots 8SM-be 9-table-LOC
 'The plates and the carrots are on the table.'

The results are not what we would expect from Adamson & Anagnostopolou's conversion and resolution processes, which would yield class 2 *ba*- agreement in (82)a,b as (83) illustrates.

(83) Predicted for (82)b *ba*; 6 out of 8 speakers choose *zi*-



Underscoring the difference between these results and those for conjoined singulars, default agreement matching neither conjunct was strongly dispreferred. Table 2 shows the results for 12 pairs of [+human] plurals of which neither is drawn from class 2. Across eight speakers, class 2 *ba*- was chosen only 10 times out of 96 examples in which it matches neither conjunct and thus is unambiguously default.⁸ The upshot is that when one plural conjunct is class 2, *ba*-agreement is more than likely formal agreement with that DP.

⁸ Interestingly in all of the cases where default *ba*- was chosen, one conjunct was class 4, which seems the "weakest" gender. I have no insights to offer but see §6.5 on a potentially relevant idiosyncrasy of class 6.

Table 3: Default agreement for combinations in which no conjunct = class 2 (eight speakers)

4+6	4+8	4+10	6+4	6+8	6+10	8+4	8+6	8+10	10+4	10+6	10+8
2	2	2	2	0	0	1	0	0	1	0	0

Recall the pattern of default agreement when singulars are conjoined, in cases like (67)a,b (reproduced below) for which I proposed the derivations in (68)a,b. In many cases including (67)a, agreement matches neither conjunct.

- (67) a. *Is-anuse nen-tombi ba-ya-sebenza.* b. *U-bhaka nen-cwadi zi-ngaphandle.*
 7-medium and.9-girl 2SM-DISJ-work 1a-backpack and.9-book 8SM-be.outside
 'The medium and the girl are working.' 'The backpack and the book are outside.'

- (68) a. $\&P$ $X\{D, E\}; \checkmark\{Gen A, Pl\} = cl\ 2\ ba$
 DP1_{sg} & DP_{sg}
 medium.D & girl.E
 A A
- b. $\&P$ $X\{1a, E\}; \checkmark\{Gen D, Pl\} = cl\ 8\ zi$
 DP1_{sg} & DP2_{sg}
 backpack.A & book.E
 D D

I conclude that gender mismatch does not trigger resolution, contra Corbett & Mtenje (1987). Only singular conjuncts do so.

6.3.2 LCA avoiding 4 and 6

The following examples show that if DP1 is class 4 or 6 and DP2 mismatches it, the preference shifts strikingly to last conjunct agreement (LCA). Agreement with first conjuncts of classes 4 and 6 is not ruled out, but it is clearly dispreferred.

- (84) *Imi-gewu na-ba-ntwana ba/i/zi-ya-cul-a.* [LCA > FCA & [-human] default 5:2:1]
 4-criminals and.2-children 2SM/4SM/8SM-DISJ-sing-FV
 'The criminals and the children are singing.'
- (85) *Ama-polisa na-ba-ntwana ba/i-ya-cul-a.* [LCA > FCA 6:2]
 6-polisa and.2-children 2SM/6SM-DISJ-sing-FV
 'The policemen and the children are singing.'
- (86) a. *Imi-nqathe nezi-tya zi-/i-se tatile-ni.* [LCA > FCA: 6:2]
 4-carrots and.8-plates 8SM/4SM-be 9-table-LOC
 'The carrots and the plate disappeared.'
- b. *Ama-qanda nezi-tya a/zi-nyamalele.* [LCA > FCA 7:1]
 6-eggs and.8-plates 6SM/8SM-disappeared
 'The eggs and the plate disappeared.'

c. Imi-funo neem-botyi zi/i-phel-ile. [LCA>FCA: 6:2]
 4-vegetables and.10-beans 10SM/4SM-be.finished-DISJ
 'The vegetables and the beans are finished.'

The greatest variability arises when classes 4 and 6 are conjoined. The plurality of my speakers chose default *zi*- and *ba*-; otherwise, FCA slightly edged out LCA:

(87) Imi-nqathe nama-qanda i-/a-/zi-se tafile-ni. [default *zi*>FCA>LCA 5:2:1]
 4-carrots and.6-eggs 4SM/6SM/8SM-be 9-table-LOC [i.e. *zi* x 5, *i* x 2, *a* x 1]
 'The carrots and the eggs are on the table.'

(88) Ama-qanda nemi-nqathe i-/a-/zi-se tafile-ni. [default *zi*>FCA or LCA 3:2:2]
 6-eggs and.4-carrots 4SM/6SM/8SM-be 9-table-LOC [i.e. *zi* x 3, *i* x 2, *a* x 2]
 'The eggs and the carrots are on the table.'

(89) Imi-gewu nama-polisa i-/a-/ba-sebenza ndawonye. [default *ba*>LCA>FCA 4:3:2]
 4-criminals and. 6-police 4SM/6SM/2SM-work together [i.e. *ba* x 4, *a* x 3, *i* x 2]
 'The criminals and the police are working together.'

6.4 Analysis of agreement with conjoined plurals

I summarize the patterns to be captured in (90).

- (90) a. Agr with gender-matching plurals is gender- and number-matching
 b. Agr with mismatched plurals is preferentially FCA *but*
 c. If DP1 bears a wholly *u*gender, the preference changes to LCA *and*
 d. Iff DP1 & DP2 both bear wholly *u*genders, results are highly variable. Default agreement is an option in this case alone, and is the plurality choice.

That a conjunction of plurals [4+4] or [6+6] yields matching *i*- or *a*- agreement and not default *ba*- or *zi*- suggests that resolution is a last resort. Adapting Marušič et al (2007) and Bošković (2009) I propose that the preferred mechanism for agreement with conjuncts is Multiple Agree, copying the plural feature of &P and the full features of both DPs. When the conjuncts are singular, mismatching &P, Multiple Agree cannot succeed. Agreement is obligatory and relies on resolution at the &P level in this case only, as detailed in preceding sections.⁹

⁹ Three points: (i) Following Bosković (2009), the fact that conjuncts must be plural is also evidence that the relevant *u*Phi probe is not a “split” probe that can obtain a number value from one goal and a gender value from another. (ii) This pattern, at odds with Preminger (2015), is likely due to labeling requirements

Let us see why we need Multiple Agree, and how it derives the results in three contrasting cases presented in the subsections above and reproduced below. (82)b and (79)a could be derived by Multiple Agree including only the first conjunct – Agree (T, &P, DP1) -- as DP1 suffices to provide the features realized in subject agreement here. But that would erroneously yield class 4 agreement in (86)c as well, where in actuality the preference is for LCA. A minimal modification could accommodate this example: if Agree (&P, DP1) applies and encounters a wholly *ugender* in DP1, probing might move on to DP2 (perhaps deleting the *ugender* of DP1). This will yield FCA in (82)b and LCA in (86)c. In (79)a, where DP2 also has *ugender* (the same one as DP1), we might suppose that as the final available option, DP2's feature is exponed.

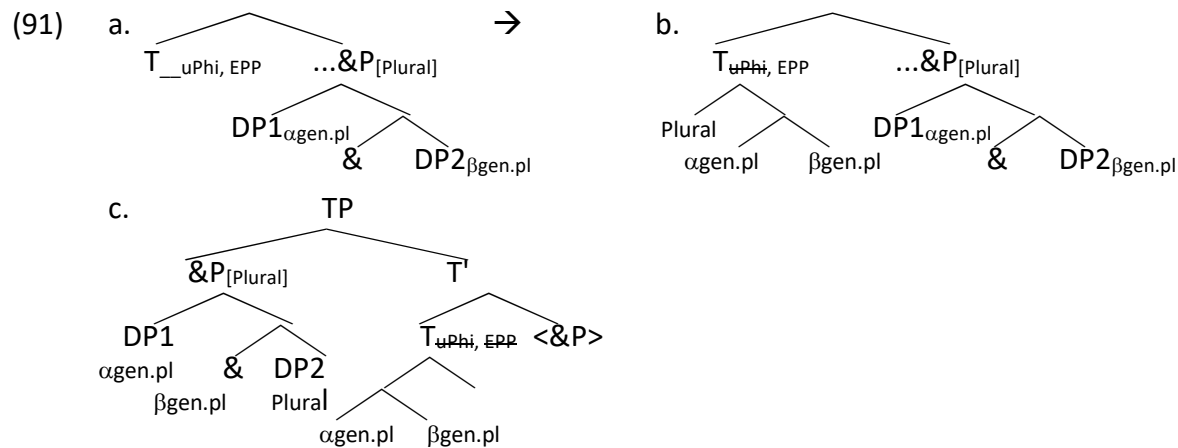
- (82) b. Iza-nuse **naba-ntwana** zi-ya-cul-a. [FCA > LCA 6:2]
 8-mediums and.2-children 8SM-DISJ-sing-FV
 'The young ladies and the children are singing.'
- (79) a. Imi-pu nemi-bhobho i-se rumi-ni. [4+4 = 4]
 4-guns and.4-pipes 4SM-LOC room-LOC
 'The guns and the pipes are in the room.'
- (86) c. Imi-funo **neem-botyi** zi/i-phel-ile. [LCA > FCA: 6:2]
 4-vegetables and.10-beans 10SM/4SM-be.finished-DISJ [4+10=10]
 'The vegetables and the beans are finished.'

But then, (87) and (88) present a puzzle: the procedure wrongly predicts preferential agreement with DP2 since DP1 has *ugender*, but instead, this is the one circumstance in which default agreement surfaces and in fact predominates.

- (87) Imi-ngathe **nama-qanda** i-/a-/zi-se tafile-ni. [default zi > FCA > LCA 5:2:1]
 4-carrots and.6-eggs 4SM/6SM/8SM-be 9-table-LOC [i.e. zi x 5, i x 2, a x 1]
 'The carrots and the eggs are on the table.' [4+6=default 8]
- (88) Ama-qanda **nemi-ngathe** i-/a-/zi-se tafile-ni. [default zi > FCA or LCA 3:2:2]
 6-eggs and.4-carrots 4SM/6SM/8SM-be 9-table-LOC [i.e. zi x 3, i x 2, a x 2]
 'The eggs and the carrots are on the table.'

(Chomsky 2013, 2015). I leave follow-up to future research. (iii) A prediction of the last resort approach to resolution, to explore in future work: if there is variation in whether &P is always plural, there should be accompanying variation in whether resolution is employed at all.

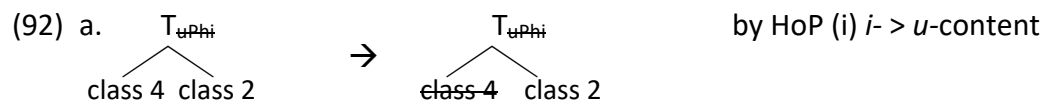
I conclude from this pattern that the features of both conjuncts factor into the choice of agreement simultaneously, not serially. The features of both conjuncts are copied to the probe establishing a reproduction of the features of the conjuncts.¹⁰ The gender features' relative left-to-right positions in the $u\Phi$ i bundles below reflects the hierarchical or linear relations among conjuncts.



Because T can expone only a single gender feature, an impoverishment operation (Bonet 1991) must delete one of them.^{11,12} The preference hierarchies in (77) determine which gender feature is retained, with (i) bleeding (ii). I assume these are language-specific; FCA is not a consistent cross-Bantu preference (Mitchley 2015).

- (77) Hierarchies of preference in agreement with preverbal conjoined plurals
 (i) Genders with *i*content > default > wholly *u*genders
 (ii) Otherwise, FCA > LCA

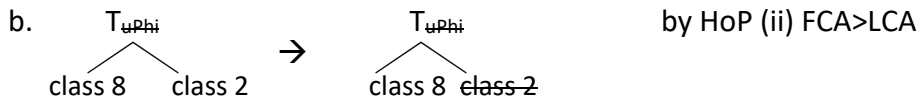
Application of the impoverishment operation is sketched in in (92).



¹⁰ See also Citko (2018) for an approach copying features of both conjuncts onto the probe and Akkus (2020) for a Multiple Agree analysis of a phi-feature dependency between subject and object in Iranian.

¹¹ Murphy & Puškar (2018) also apply impoverishment to derive the features of agreement with conjuncts. In their account, impoverishment reduces multiple gender values on &P itself, to arrive at the least marked.

¹² For the speakers that I consulted, conjunctions of more than two DPs always pair with default/semantic agreement. This might indicate a limitation of the Multiple Agree process; I leave it to future research.



When default agreement accompanies conjoined plurals like (87)-(89) I suggest that there is deletion of both gender values on the probe, rendering its *ugender* unvalued. Default insertion then must take place.

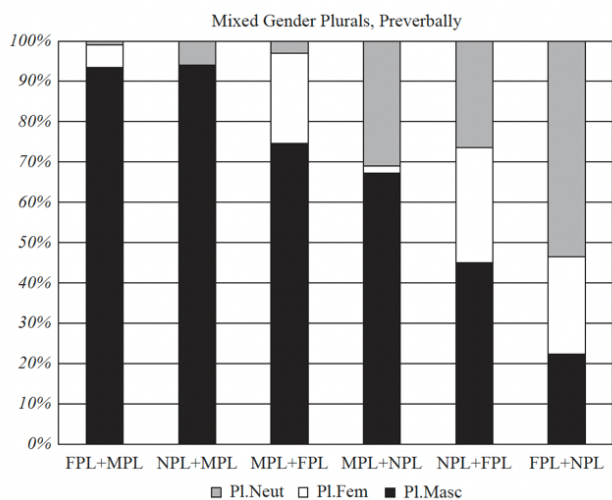


These strategies do not work for conjoined singulars since Multiple Agree fails for $*(T, \&P_{pl}, DP1_{sg}, DP2_{sg})$ as we have seen, so only percolation and resolution can value the probe.

6.5 Conjoined plurals in BCS and existing approaches

It is already well-established that gender agreement with conjoined plurals varies across speakers. Table 3 represents responses on different combinations in Slovenian.

Table 3 Slovenian conjoined plurals (Marušič et al 2015)



There are a number of approaches to the variability of judgments on conjoined plurals.

Marušič et al (2015), Marušič & Nevins (2020) propose a Distributed Agree model under which agreement applies in two stages, Agree Link and Agree Copy. If Agree Copy applies before linearization, the result is highest conjunct agreement; if Agree Copy applies after linearization

then there is closest conjunct agreement because the structure is flattened out. I represent this schematically in (94) and (95).

- (94) Agree-Copy before linearization: default gender or agreement with DP1, regardless of word order, because what's visible to syntax is hierarchical structure



- (95) Agree-Copy after linearization: default gender or closest conjunct agreement, because hierarchical structure ceases to be visible



This approach seems to assume random results, but as we have seen, agreement with conjoined plurals in Xhosa is variable but not random. To capture the Xhosa hierarchies of preference for conjoined plurals would require massive look-ahead establishing, for instance, that DP2 is more agreeable than DP1, making it preferable to wait and do Agree Copy after linearizing.

A similar problem arises in Murphy & Puškar (2018), who propose that the head & can in principle obtain multiple gender values from its conjuncts by Agreeing with them. But which values it acquires depends on the order of application among the operations Merge, Agree Up, and Agree Down.

If Agree Up applies before Merge, and then Agree Down applies after Merge, there will be LCA. The reason for this is that with this order of operations, Agree Up applies vacuously before the first conjunct is present. Agree Down will successfully give &P the gender feature of its second (lower) conjunct, so the result must be LCA. Agree Down preceding Merge with Agree

Up applying last yields FCA, because in this case Agree Down is vacuous, but Agree Up is successful. Hence & has features of the first conjunct and agreement must be FCA.

Like Distributed Agree, this provides no handle on the way both conjuncts' gender features impact agreement with conjuncts in Xhosa. And in fact, the Slovenian results depicted in Table 4 are also variable, but not random. When there is a masculine conjunct, masculine agreement prevails. And for any given gender, the odds of its determining agreement are better if it is borne by the second conjunct than by the first. Table 4 shows that anytime a conjunct is masculine it is the most likely one to value agreement, especially if it is the final conjunct. Neuter is more likely to be adopted than either feminine or the masculine default, when neuter and feminine are combined. Feminine is more likely to be chosen when it's the last conjunct than when it is the first, but it is the least likely overall. Something along the lines of (96) is motivated:

- (96) Draft hierarchy of preferences for agreement with conjoined plurals in Slovenian
- (i) Masc > neuter > default > feminine
 - (ii) LCA > FCA

It lies outside the scope of this brief section to delve further into these preferences and how they interact. The point I wish to make is that trends are detectible for Slovenian in Table 4 as well as in the Xhosa data that we've explored. These cannot be captured in an approach designed to generate randomly variable results.

6.6 For future research

Before leaving conjoined plurals, I describe a loose end that I must regretfully leave unaccounted for, having to do with [human] nouns of plural class 6.

When DP1 is a [human] noun of class 6 and DP2 is of class 8 or 10, FCA predominates, unlike in the case of an [inanimate] class 6 noun; the latter patterns along with class 4 to shift the preference to LCA under (77)(i). The pattern for a [human] DP1 of class 6 is more mixed than when classes 2, 8, or 10 are DP1, but nevertheless it favors FCA. (86) (repeated below) illustrates the contrasting pattern for an [inanimate] DP1 of class 6 or 4.

- (97) a. Ama-gqirha nez-anuse a-/ba/zi-sebenz-a ndawonye. [FCA>default>LCA]
 6-doctors and.8diviners 6SA/2SA/8SA-work-FV together
 'The traditional doctors and the diviners are working together.'
 [results: FCA x 5, default *ba-* x 1, FCA or default x1, LCA x 1]
- b. Ama-polisa neen-tombi a-ya-cul-a. [FCA>LCA>default]
 6-police and.10-girls 6SA/-DISJ-sing-FV
 'The policemen and the young ladies are singing.'
 [results: FCA x 4, LCA x 2, FCA or LCA x 1, default *ba-* x 1]
- (86) a. Imi-nqathe **nezi-tya** zi-/i-se tatile-ni. [LCA >FCA: 6:2]
 4-carrots and.8-plates 8SM/4SM-be 9-table-LOC
 'The carrots and the plate disappeared.'
- b. Ama-qanda **nezi-tya** a/zi-nyamalele. [LCA >FCA 7:1]
 6-eggs and.8-plates 6SM/8SM-disappeared
 'The eggs and the plate disappeared.'
- c. Imi-funo **neem-botyi** zi/i-phel-ile. [LCA>FCA: 6:2]
 4-vegetables and.10-beans 10SM/4SM-be.finished-DISJ
 'The vegetables and the beans are finished.'

In all other classes -- 1/2, 3/4, 7/8, 9/10 -- [human], [animal], and [inanimate] nouns pattern alike; this fact led to my proposal that *u-* vs. *i-* flavors of a gender are indistinguishable in the syntax. What can be the source of this exceptional behavior of class 6? A correlate provides a clue: class 6 serves as plural for many [human] nouns whose singulars fall in other classes.¹³

- (98) a. in-doda/ama-doda b. in-kosi/ama-khosi c. in-kosilazi/ama-kosilazi
 9-man/6-man 9-king/6-king 9-wife/6-wives
 'man/men' 'king/kings' 'wife/wives'
- d. um-Xhosa/ama-Xhosa e. um-Zulu/ama-Zulu
 1-Xhosa/6-Xhosa 1-Zulu/6-Zulu
 Xhosa person/Xhosa people or nation' 'Zulu person/Zulu people or nation'

¹³ In some Bantu languages class 6 is a generic plural, combining with nouns of any semantics and classes; the Xhosa pattern is more restrictive.

The translations in (98)d and (98)e reflect a kind of ‘group’ reading often associated with class 6 plurals of [human] nouns. I propose that these plurals correspond to a distinct gender F which has the meaning ‘group’ underlying a prefix and agreement class that I’ll call 6x, homophonous with the class 6 plural of gender C but combining only with a marked subset of [human] nouns.

(99) ns for Xhosa, reflecting conjunct agreement facts

Classes 1/2 = Gender A	<i>n_{A1}</i>	<i>i</i> [human]
	<i>n_{A2}</i>	<i>arb</i>
Classes 3/4 = Gender B	<i>n_B</i>	uninterpretable, for all members
Classes 5/6 = Gender C	<i>n_C</i>	uninterpretable, for all members
Classes 7/8 = Gender D	<i>n_{D1}</i>	<i>i</i> [artifact]
	<i>n_{D2}</i>	<i>arb</i>
Classes 9/10 = Gender E	<i>n_{E1}</i>	<i>i</i> [animal]
	<i>n_{E2}</i>	<i>arb</i>
Class 6x = Gender F	<i>n_F</i>	<i>i</i> [human group]

Because it is interpretable, gender F/6x falls under preference (i) of (77). But [human] nouns pluralized with *ama-* are ambiguous, leaving the door open to some variability.

This leaves (85) to be explained: why is FCA not the choice of four or five speakers, as it is in (97) where DP2 is class 8 or 10? Instead, LCA is favored.

(85) Ama-polisa na-**ba-ntwana** **ba**/i-ya-cul-a. [LCA > FCA 6:2]
 6-polisa and. 2-children 2SM / 6SM-DISJ-sing-FV
 ‘The policemen and the children are singing.’

It seems that class 2, with its consistent status as interpretable gender and semantic agreement, outranks the ambiguously (un)interpretable class 6. I leave it to future research to find an insightful way to integrate this into the account of agreement with conjoined plurals.

7 Agreement failure and exceptionality

7.1 Agreement and special semantics

This section considers the question of what precisely “uninterpretable” means in relation to classes 3/4 and 5/6: do they really have no semantic content, contra Denny & Creider (1986), or

rather do they have content incompatible with the grammar of agreement with conjoined singulars? Building on Arsenijevic (2017, 2021), Adamson & Anagnostopoulou (2024) relate the failure of neuter plural agreement with conjunctions of singular neuters in BCS to the fact that many neuters are mass nouns, lacking individuation and countability. It is worth exploring whether something along these lines could be true of classes 3/4 and 5/6 in Xhosa, based on Denny & Creider's proposed meanings of solid figure/(non-)extended shape for these pairs.

Supporting evidence turns out to be elusive. I could find no symptoms of "outlier" properties in 3/4 and 5/6, apart from the failure of gender-appropriate plural agreement with gender-matched conjoined singulars that it is my goal to derive. Classes 3 and 5 are not among those lacking plurals or devoted to abstracts; the nouns in these classes are mostly ordinary objects that are countable and that pluralize regularly as shown in (100).

- (100) a. **um**-thi **omu**-nye b. **imi**-thi **imi**-thathu c. **i**-qanda **eli**-nye d. **ama**-qanda **ama**-thathu
 3-tree 3-one 4-trees 4-three 5-egg 5-one 6-eggs 6-three
 'one tree' 'three trees' 'one egg' 'three eggs'

Apart from class 1, every Xhosa class includes some nouns referring to uncountable substances such as liquids and/or masses as I showed in (15), repeated below.

(15) Liquids and masses: scattered

- | | | | | | |
|---|--|--|---|-------------------------------------|--|
| a. um -chamo
3-urine
'urine' | b. i -gazi
5-blood
'blood' | c. ama -nzi
6-water
'water' | d. isi -dudu
7-porridge
'porridge' | e. i -oili
9-oil
'oil' | f. u -bisi
11-milk
'milk' |
| g. um -gubo
3-flour
'flour' | h. ama -zimba
6-sorghum
'sorghum' | i. i -tyuwa
9-salt
'salt' | j. u -daka
11-mud
'mud' | | |

These nouns must combine with a unit counter for enumeration to be possible as shown in (101)-(102). But (101)c shows that this problem arises for a liquid in class 9 – a class in which conjoined singulars take regular plural agreement. Moreover, my consultants offered as the counter for liquids *ithonzi* - 'drop' in (102), itself a noun of the problematical class 5. Taken

together this and the examples in (100) argue strongly against any deficiency in countability correlating with the need for default agreement with conjoined singulars of classes 3 and 5.

- (101) a. in-cwadi e-nye b. in-tombazana e-nye c. #i-oili e-nye
 9-book 9-one 9-girl 9-one 9-oil 9-one
 'one book' 'one girl' Literally: one oil
- (102) a. #i-gazi eli-nye b. i-thonzi (eli-nye) l-e-gazi / l-e-oili
 5-blood 5-one 5-drop 5-one 5-of.5-blood. /5-of.9-oil
 Literally: one blood' 'a/one drop of blood/oil' [unit counter is cl.5]

Arsenijevic (2017, 2021) provides an additional diagnostic for defectiveness of the nouns in neuter gender: he reports that it is unavailable in "highly individuated plural uses such as....in demonstratives with pointing..." (104) from Arsenijevic (2021:21). (104)a-d show that this is not a problem for classes 5/6 and 3/4, whether singular or plural.

- (103) Ona *(deca) su neumorna. (pointing at children)
 those.NEUT.PL child. NEUT.PL AUX.PL tireless.NEUT.PL
 'Those children are tireless'/*'Those are tireless.'
- (104) a. Loo (m-thi /m-gulukudu) m-de. b. Eli (qanda/gqwetha) l-awa.
 3that 3-tree/3-gangster 3-tall 5that 5egg/5lawyer 5sm-fell
 'That (tree/gangster) is tall' 'That (egg/lawyer) fell'
- c. Le (mi-thi /mi-gulukudu) im-bi. d. La (ma-qanda/ma-nenekazi) ma-hle.
 4those (4-trees/4-gangsters) 4-bad 6those (6-eggs/6-ladies) 6-nice
 'Those (trees/gangsters) are bad.' 'Those (eggs/ladies) are nice.'

Pronominal reference and pronoun binding are also relevant tests. In certain other languages, for nouns whose grammatical gender is at odds with their semantic gender, it is possible to utilize pronouns that match the latter rather than the former ((105) from Mann 1979 via Braun & Haig 2010; glosses added). This phenomenon is frequently used evidence that neuter is special.

- (105) Das junge Mädchen gab ihm die Hand. Sie wollte wohl nett sein,
 the young girl_{neut} shook him the hand. she_{fem} wanted probably nice be
 aber was war mit ihr anzufangen.
 but what was with her_{fem} start
 'The young girl shook his hand. Apparently, she intended to be nice, but what could be done with her.'

In Xhosa however this pattern is not replicated -- in fact, noun class matching is very strict; non-matches yield disjoint readings. (106) is a discourse in which subject and object markers match a class 5 human-denoting noun and (107) is parallel discourse involving class 10. In neither case can a pronoun or agreement marker of canonically human classes 1/2 be substituted.

- (106) I-gwetha I-asondela kwi-jury. L- /#u-ashwankathela i-tyala.
 5-lawyer 5SM-approach LOC.9-jury 5SM-/1SM-summarized 5-case
 'The lawyer approached the jury. He summarized the case.
 Ba-li-/#m-phulaphula ngenyameko.
 2SM-5OM-/#1OM listen carefully
 They listened carefully to him.'
- (107) lin-tombi z-a-thenga ama-hashe. Z- a- zi-nga-kwazi uku-wa-khwela.
 10-girls 10SM-buy 6-horses 10SM-PST-10SM-NEG-know 15-6OM-ride
 Aba-hlobo ba-zo /#bo ba- be- zi- /#ba-hleka.
 2-friends 2-10POSS-/2POSS 2SM-PST-10OM/#2OM-laugh
 'The girls bought horses. They didn't know how to ride them.
 Their friends laughed at them.'

Bound readings for Xhosa pronouns also require noun class matching, in every class; in contrast, a semantically feminine but formally neuter noun in Greek can bind either a neuter or feminine pronoun (see (109) from Spathas 2010:222).

- (108) a. I-nenekazi nga-li-nye_i I-a-yi-funda in-cwadi ya-lo_i /ya-khe_j. *_i = _j
 5-lady each-5-one 5SM-PST-9OM 9-book 9-5POSS /9-1POSS
 'Each lady_i read her_i book.'
- b. Y-onke in-kwenkwe_i y-a-tya imi-funo ya-yo_i /ya-khe_j. *_i = _j
 9-every 9-boy 9SM-PST-eat 4-vegetables 4-9POSS /4-1POSS
 'Every boy ate his vegetables.'
- c. Y-onke imi-gulukudu_i y-a-khwela ama-hashe a-yo_i /a-bo_j. *_i = _j
 4-all 4-gangster 4SM-PST-ride 6-horses 6-4POSS /6-2POSS
 'All the gangsters rode their horses.'
- (109) Kathe koritsi diakosmise to dhomatio {tis/tu}
 every girl.N decorated the room hers/its
 'Every girl decorated her room.'

I conclude that Xhosa genders 3/4 and 5/6 display no symptoms of defectiveness apart from the failure of matching agreement with [3+3], [5+5].

There is, I think, a lingering possibility that semantic content to these genders lies at the root of the agreement problem. Ordinary plural formation is accomplished through a Number head with a plural value as shown in (9) (Ritter 1991, Carstens 1991), and as for a noun's gender, there is no competition or obstacle generally for its association with its own containing DP. In contrast, the plurality and gender of &P dominating conjoined singulars is determined in a completely different way – I've adopted the view that &P has a plural feature, but as Corbett & Mtenje point out and as we have seen, singular conjuncts trigger the resolution process; I think this tells us that their number features are in some sense added together by the grammar as part of Adamson & Anagnostopolou's *conversion*. We know little enough about this that I think it is in principle possible that non-canonical semantic features interfere.

As for the failure of the defectivity tests, it's potentially relevant that coreference and variable binding are the domain of so-called index features (Wechsler & Zlatic 2000, 2003). In contrast the resolution process operates on concord features – number and gender in isolation. The tests, while useful in establishing some properties of neuter, may not be getting at the right properties where the resolution process is concerned.

All in all, though, I think we must conclude that the hypothesis fails, at least based on what we presently know. That semantic features associated with certain genders interfere in resolution is an unfalsifiable conjecture if there are no reliable diagnostics. The properties of neuter identified as telltale by Arsenijevic (2017, 2021), Adamson & Anagnostopolou (2024) have no parallels in Xhosa, so it isn't clear whether they are relevant or coincidental even in the case of neuter. For practical purposes, these genders are uninterpretable: they have no content that is usable in the resolution calculus, and are ignored.

7.2 Evidence from Shona

In Xhosa we've seen that gender-matching conjoined singulars pair with matching plural agreement in three out of five of the genders considered; [3+3] and [5+5] are the sole exceptions. In BCS, singulars from two out of three genders pair with expected agreement; neuter is the sole odd man out. These patterns give the impression that matching agreement is the norm and its failure the exception to be explained. But the Shona language raises a question regarding the generality of this. In Shona, the genders pairing with default agreement outnumber those in which matching agreement succeeds. The facts suggest that genders which *do* control matching agreement are the exceptional ones, related to a very small number of semantic primitives; these alone are interpretable gender features, for purposes of resolution. Much about Shona agreement parallels what we have seen in Xhosa: conjunctions [1+1] and [7+7] pair with matching agreement. I have not encountered non-human-denoting nouns in Shona class 1, but there are both [human] and [inanimate] nouns in class 7 and all pair with expected plural agreement of class 8 when conjoined (see (110)-(112)).

- (110) Mu-rume ne mu-kadzi **va**-ri panze. [Shona: 1+1=2]
1-man and 1-woman 2SM-be outside
'The man and the woman are outside.'
- (111) Chi-ngwa ne chi-bage **zvi**-ri pa-tafura. [Shona: 7+7=8]
7-bread and 7-maize 8SM-be LOC-table
'The bread and the maize are on the table.'
- (112) Chi-dhakwa chi-refu nechi-dkakwa chi-pfupi zvi-ri ku-netsana [Shona: 7+7=8]
7-drunkard. 7-tall and 7-drunkard 7-short. 8SM-be 15argue
'The tall drunkard and the short drunkard argued.'

[Human] and [inanimate] nouns of [3+3] and [5+5] pair with agreement of classes 2 and 8 respectively, just as in Xhosa:

- (113) Mu-goti ne mu-ti Xi /**zvi**-ri panze. [Shona: 3+3=8]
3-cooking.stick and 3-tree 4SM/ 8SM-be outside
'The cooking stick and the tree are outside.'

(114) Dombo ne zai Xa /✓**zvi**-ri panze. [Shona: 5+5=8]
 5stone and 5egg 6SM/ 8SM-be outside
 'The stone and the egg are outside.'

(115) Benzi ne dinga Xa- /✓**va**-ri ku-shayikwa. [Shona: 5+5=2]
 5fool and 5dimwit 6SM/ 2SM-be 15-missing
 'The fool and the dimwit are missing.'

Shona has eight regular singular/plural pairs of noun classes – three more than Xhosa, plus a few singletons. Yet strikingly, only [1+1] and [7+7] pair with the regular plural agreement of their genders. The rest pattern with [3+3] and [5+5] in taking only default plurals. [9+9] nouns do not pair with class 10 agreement like they do in Xhosa. This is easily determined because class 8 and 10 agreement are distinct in Shona, *zi-* vs. *dzi-*, unlike in Xhosa where both are *zi-*.¹⁴ In Shona, class 9 nouns show the pattern illustrated for class 5 in (114) and (115): [human] nouns take class 2 *va*-agreement as shown in (116)a, and the rest – animals, body parts, artifacts and abstracts – pair with class 8.

- (116) a. N-herera ne n-yanzvi **va**-/*dz-aka-on-ana.
 9-orphan and 9-expert 2SM-/*10SM-PST-see-RECIP
 'The orphan and the expert saw each other.'
- b. N-herera dz-aka-on-ana.
 10-orphans 10SM-PST-see-RECIP
 'The orphans saw each other.'
- c. Im-bwa nhema nem-hou chena **zv**- /*dz-aka-rwa.
 9-dog 9-black and 9-cow 9-white 8SM-/*10SM-PST-fall
 'The black dog and the white cow fell.'
- d. M-huno iyi ne nzeve iyi **zvi**- /*dz-ne utachiona.
 9-nose 9-this and 9-ear 9-this 8SM-/*10SM-have 14-infection
 'This nose and this ear are infected.'
- e. M-biya iyi ne sando iyo **zvi**- /*dz-ne n-gura.
 9-bowl 9-this and 9-hammer 9-that 8SM-/*10SM-have 9-rust
 'This bowl and that hammer are rusty.'
- f. N-yota ne n-zara **zv**- /*dz-aka-tambudza va-fambi.
 9-hunger and 9-thirst 8SM-/*10SM-afflict 2-travelers
 'Hunger and thirst afflicted the travelers.'

¹⁴ It is not the case that syncretism in Xhosa camouflages a similar pattern, since recall that both [human] and [inanimate] nouns of Xhosa class 9 combine with *zi-* when conjoined, just as all [1+1] pair with 2agr and [7+7] pair with 8agr.

Shona has class 11 nouns which pluralize in class 10. Like [9+9], conjoined [11+11] nouns combine with default class 8 *zi-* and not with class 10 *dzi-* which agrees with their plurals:

- (117) Ru-oko noru-rimi zvi-/*dzi-ne u-tachiona.
 11-hand and.11-tongue 8SM-/*10SM-have 14-infection
 'The hand and the tongue are infected.'

Nouns of class 14 pluralize in class 6, but conjoined class 14 singulars pair with class 8 *zi-*, not with the *a-* subject marker that goes with their plurals (see 0a,b):

- (118) a. U-ta ne u-tanho zvi-ri/*a-ri kunze.
 14-bow and 14-ladder 8SM-/*6SM-be outside
 'The bow and the ladder are outside.'
- b. Ma-uta ne ma-utanho a-ri kunze.
 6-bows and 6-ladders 6SM-be outside
 'The bows and the ladders are outside.'

Diminutives are formed with the prefixes *ka-/twa-* of classes 12/13. Their plurals pair with *twa-* agreement (see (119)a,b). But (120) shows that when diminutives conjoin, agreement must be taken from class 2 or 8.

- (119) a. mu-sikana → ka-sikana b. Twa-sikana twa-nyangadika.
 1girl 12-girl 13-girls 13SM-disappear
 'girl' 'tiny girl' 'The tiny girls disappeared.'
- (120) a. Ka-sikana ne ka-kómáná va-/*twa-nyangadika.
 12-girl & 12-boy 2SM/*12-disappear
 'The tiny girl and the tiny boy disappeared.'
- b. Ka-mba ne ka-motokari zva-/*twa-nyangadika.
 12-house & 12-car 8SM/*13SM-disappear
 'The tiny house and the tiny car disappeared.'

I suggest that the diminutive morpheme is a silent root, pairing with a gender that is wholly uninterpretable as shown in (121) (see Creemers & Fenger 2018, Lowenstamm 2015, Simonović 2022, Wiltschko & Steriopolo 2007 on (some) derivational affixes as roots.) This is better compatible with the facts of agreement in (120) than is attribution of the diminutive meaning to class 12/13, or to a dedicated *n* of that gender (Kramer 2015, Fuchs & Van der Wal 2022).

- (121) a. diminutive structure:
-
- b.
-

Nonetheless, conjunctions of gender-matched plurals pair with matching plural agreement as we saw to be true of all plural coordinations in Xhosa; the pattern holds in Shona as well.

- (122) a. Twa-sikana ne twa-komana twa-/*va-nyangadika.
 13-girl and 13-boy 13 /*2_{SM}-disappear
 'The tiny girls and the tiny boys disappeared.'
- b. M-huno idzi ne nzeve idzodzo dzi- /*zvi-ne utachiona.
 10-nose 10these and 10ear 10those 10_{SM}-/*8_{SM}-have 14-infection
 'These noses and those ears are infected.'

Summing up, Shona has nearly three times the number of genders as better-studied European languages, and several more than the five main singular/plural pairings found in Xhosa. But of these, only two show regular plural agreement with conjoined singulars. This raises the possibility that it is the latter pattern that is exceptional, perhaps because the number of genders with clear semantic associations is typically small.

Dechaine et al (2014) argue that Shona class prefixes are multi-functional, having intrinsic semantics and derivational functions as well as their singular/plural formation functions; these are summarized in Table 4 below. See however Carstens (1991, 2008) for proposals that the apparent derivational functions of Bantu noun classes are actually performed by null derivational morphemes of the relevant genders as in the sketch of diminutive (121).

Table 4 (Dechaine et al 2014: 25)

Descriptive and evaluative denotations of Shona noun-class prefixes (shaded cells are multi-functional).

PREFIX	DESCRIPTIVE DENOTATION		EVALUATIVE DENOTATION			
SG	<i>mù-</i>	<i>mù₁kómáná₁</i>	‘boy’	—	—	
PL	<i>và-</i>	<i>và₂kómáná₁</i>	‘boys’	HON	<i>và_{HON}mù₁kómáná₁</i>	‘Mr/Ms Boy’
	<i>vá-</i>	—	—		<i>vánávà_{HON}mù₁kómáná₁</i>	‘Mr/Ms Boys’
SG	<i>mù-</i>	<i>mù₃tí₃</i>	‘tree’	—	—	—
PL	<i>mì-</i>	<i>mì₄tí₃</i>	‘trees’	—	—	—
SG	<i>VCE-</i>	<i>VCE₅gòré₅</i>	‘cloud’	AUG	<i>VCE₅gómáná₁</i>	‘big boy’
PL	<i>mà-</i>	<i>mà₆kòré₅</i>	‘clouds’		<i>mà₆VCE₅gómáná₁</i>	‘big boys’
SG	<i>chì-</i>	<i>chì₇nhù₇</i>	‘thing’	DIM	<i>chì₇kómáná₁</i>	‘small sturdy boy’
PL	<i>zví-</i>	<i>zví₈nhù₇</i>	‘things’		<i>zví₈kómáná₁</i>	‘small sturdy boys’
SG	N-	<i>N_{9/10}shùmbá₉</i>	‘lion(s)’	—	—	—
PL	<i>rù-</i>	<i>rù₁₁kòvá₁₁</i>	‘stream’	DIM, PEJ	<i>rù₁₁kómáná₁</i>	‘thin scraggly boy(s)’
PL	N-	<i>N₁₀hòvá₁₁</i>	‘streams’	—	—	—
SG	<i>kà-</i>	<i>kà₁₂mbùnyú₁₂</i>	‘insect’	DIM	<i>kà₁₂kómáná₁</i>	‘tiny boy’
PL	<i>tù-</i>	<i>tù₁₃mbùnyú₁₂</i>	‘insects’		<i>tù₁₃kómáná₁</i>	‘tiny boys’
SG	<i>ù-</i>	<i>ù₁₄swá₁₄</i>	‘grass’	—	—	—
PL		<i>mà₆ù₁₄swá₁₄</i>	‘grasses’			
SG	—	—	—	AUG	<i>zì₂₁VCE₅gómáná₁</i>	‘big boys’
PL	—	—	—		<i>mà₆zì₂₁VCE₅gómáná₁</i>	‘big boys’

As noted in §7.1, diagnostics are lacking to link the proposed semantic associations to genders and to the difficulty of resolution for conjoined singulars. It is perhaps worth noting from this table that Shona classes 7/8 have a diminutive denotation when added to nouns whose intrinsic classes are other than 7/8; yet 7/8 nouns do take regular class 8 agreement when pluralized, unlike nouns of the dedicated diminutive class 12. Diminutive content thus cannot be what blocks agreement with conjoined class 12 singulars.

8 Conclusions

I have argued that a modest but grammatically significant semantic core underlies the Bantu noun class system: [human], [inanimate], and in Xhosa at least, [animal]. So-called default agreement with conjoined singulars in Bantu is actually syntactic agreement with an igender core beneath wholly *ugenders* including 3/4, 5/6, Shona class 12/13, and a number of others.

Singular nouns from only a small number of genders pair with gender-matching plural agreement when conjoined, and these genders have semantic associations. For Xhosa classes 1/2 and 7/8 the associations are clear from their role in default/semantic agreement.

The phenomenon gives rise to a predictable correlation:

- (123) Gender α has interpretable content $\rightarrow$ [$\&P$ DP1.sg_{gender. α} & DP2.sg_{gender. α}] = gender α .pl agr
 Gender α is wholly uninterpretable $\rightarrow$ [$\&P$ DP1.sg_{gender. α} & DP2.sg_{gender. α}] = default agr

By this diagnostic, it is uninterpretable of the gender that leads to failure of agreement with conjunctions of BCS and Slovenian neuter.

I have also argued that different flavors of a given gender are indistinguishable in syntax because their pairings with roots are regulated by semantic licensing conditions which apply at LF. This is a departure from Kramer (2015) where it is proposed that only semantic licensing conditions apply in LF.

I have presented evidence that only conjoined singulars trigger resolution on $\&P$, because it is the last resort approach to providing agreement with conjuncts. Where possible Multiple Agree values the probe, but if $\&P$ is plural this works only for number-matching conjoined plurals.

Though I have adopted Adamson & Anagnostopoulou's approach to agreement resolution for conjoined singulars, an important aspect of it does not play a role in my account: feature hierarchies and the dependencies among features within them, as discussed briefly in §5.

Relatedly, the Bantu patterns differ from those of the three-gender systems that Adamson & Anagnostopoulou analyze in the fact that [human] functions as a default. Adamson & Anagnostopoulou argue for the generalization in (124), which clearly does not hold in Bantu. I

argued that Bantu's resolved agreement is not determined by working from most to least specific along a feature hierarchy, based on which the irrelevance of (124) follows naturally.

(124) **Animacy Generalization:** genders linked to non-animate reference are featurally less specific...morphosyntactic default environments can give rise to exponence that realizes the features of the non-animate gender, but not the features that realize the animate...(Adamson & Anagnostopoulou 2024 on defaults in 3-gender languages)

Summing up, it does not seem that the kind of hierarchies which determine agreement with conjuncts in these languages play a role in Bantu. Instead, I have argued that default values are arrived at by peeling away layers that overlay semantic primitives.

I did find some echoes of defaults working from greater to lesser specificity in a brief preliminary exploration of agreement with conjoined locativized nouns of Chichewa. Of these noun classes, 18 *mu* is locations 'in', 17 *ku-* is locations 'at', and 16 *pa-* is locations 'on' as shown below.

- (125) a. Mu-nyumba mu-ku-nunkha. (Carstens 1997:362)
 18-9house 18-asp-stink
 'Inside the house stinks.'
- b. Ku-nyumba 'ku ndi ku-tali
 17-9-house 17dem cop 17-far
 'That house and its environs are far away.'
- c. pa-li nchenche pa ch-iseko. chi-ri-ch-onse (Carstens 1997:384)
 16SA-be 10fly 16LOC 7-door 7-be-7-all
 'There are flies on every door.'

Conjoined locatives of the same class control matching agreement as shown in (126), but when they mismatch, resolved agreement is class 16 *pa* even if neither conjunct is *pa*-prefixed, as in (127)c. My speaker reported that *pa* is needed 'for multiple places,' but it is not plural; rather, though this isn't obvious from the translations, it seems to function as the more general locative.

- (126) a. Mu nyumba ndi mu-galimoto *ku-/ *pa-/mu-na-nunkha
 18-9house and 18-9car 18- /16- /18SM-ASP-stink
 'Inside the house and inside the car stink.'

- b. Pa-/*mu-/*ku-li ngalawa pa-nyanja ndi pa udzu
16SM-/18-/17-be 10boats 16-lake and 16-grass
'There are goats on the lake and on the grass.'
- (127) a. Mu-nyumba ndi pa-bwalo *mu-/*ku-/pa-ku-nunkha.
18-9house and 16-5yard 18-/ 17-/16SM-ASP-stink
'The inside of the house and the yard stink.'
- b. Pa-nyumba ndi ku-nyanja *mu / *ku-/ pa-ku-yaka.
16-9house and 17-9lake 18-/ 17-/16SM-ASP-burn
'Fire is burning at the house and at the lake.'
 - c. Ku-sukulu ndi mu-nyumba *mu / *ku-/pa-ku-nunkha.
17-9school and 18-9house 18-/ 17-/16SM-ASP-stink
'It stinks at school and in the house.'

I doubt that there is need to view this in terms of a fixed semantic hierarchy from more to less specific places, however; rather, it seems an example of finding the common denominator in sorting things. This strikes me as a third factor process which may also underlie conversion in Adamson & Anagnostopolou's model.

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